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**Progettazione di un'ontologia giuridica per i requisiti di licenza nel diritto del credito: un caso di studio per il controllo della conformità rispetto all'articolo 29 della legge australiana sulla tutela del credito al consumo del 2009**

(In English)

# Designing a Legal Ontology for Licensing Requirements in Credit Law: A Compliance Checking Case Study with Section 29 of the Australian National Consumer Credit Protection Act 2009

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Anno Accademico 2025/2026

I hereby declare that this thesis is my own work and has been completed independently. To the best of my knowledge and belief, it contains no material previously published or written by another person, except where proper acknowledgment has been made in the text. This work has not been submitted for any other degree or diploma at this or any other institution

**Signature**_________________

**Hassan, Syed Bilal**

# DEDICATION

Foremost, I want to offer this endeavor to My Dear **Almighty ALLAH** for the wisdom, strength, knowledge, peace of mind, and good health He bestowed to go on with this research.

To my Late Father **Syed Ijaz Hussain Shah**, whose memories motivates me to reach higher, and for my mother, whose unwavering care and strength have been my anchor throughout life. This is all for you, **Ammi & Abu**,I hope I made you proud.

To my **Siblings**, whose love, encouragement, and presence have been my constant strength throughout this journey.

With love, respect, and gratitude, I dedicate this thesis to My Beautiful life partner, **Sartaaj**, and Cherished Son, JIBRAN, Who came into my life and filled it with joy, love, and purpose, inspiring me to achieve success.

# About The Team:

## Researcher:

**Syed Bilal Hassan** is a master's student at the University of Florence, specializing in information systems and semantic technologies. His research focuses on the design and implementation of legal ontologies for compliance checking, particularly in the domain of credit licensing regulations. In this project, he developed the ontology model, reasoning framework, and web-based interface.

## Supervisor:

**Enrico Francesconi** is a Research Director at the National Research Council of Italy (CNR) and a faculty member at the University of Florence. His research focuses on legal informatics, semantic web technologies, and knowledge representation, particularly in the area of legal compliance and regulatory modeling. He supervised and guided the development of this research project.

# Abstract

Legal compliance is highly regulated modeled and formalized such as credit law often a requires careful interpretation of rules and conditions. In practice, this process is largely manually, making it time consuming and potentially inconsistent. This thesis explores and demonstrates how such challenges can be addressed and expressed through the use of Semantic technologies, by transforming legal provisions and legal restrictions into a structured and modeled machine interpretable form.

Focusing on section 29 of the Australian National Consumer Credit Protection Act 2009, This study develops and express a legal ontology that reflects and captures the key core concepts underlying licensing requirements in credit activities in the financial markets. Core elements such as Agents, Licenses, Credit Activities, Principals, and Legal defenses are modeled using the Web Ontology Language (OWL), allowing Legal conditions and legal checks through formal relationships and logical constraints. This structure representation reduces ambiguity and expresses a clear interpretation of the legal text.

Molding and building on fundamental concepts and on this foundation, this thesis demonstrates how automated compliance checking can be achieved through ontology reasoning by applying logical inference, the system is able to determine whether given agent satisfies the checks the required legal conditions and conditions of compliance or falls within non-compliant scenarios. The ontology is further supported by a triple store for data management and SPARQL for retrieving and inferencing and analyzing results, forming a complete and functional framework.

To accountability and validity of this approach, a real time case study inspired by real world regulatory enforcement is presented. Different conditions and different scenarios are examined including license agents and unlicensed agents and agents acting on behalf of licensed principals and other situations and scenarios involving legal defenses. The result show that the system which is a reasoning system can consistently and accurately infer compliance outcomes in line with the legal provisions in a very simpler and accessible way.

Moreover, in addition to the back-end ontology and reasoning components, a simpler web-based user interface (UI) was developed to facilitate and express the interaction within the system to the non-technical users for the better understanding. The interface allows us to input relevant information related to agents licenses and credit activities in a structured and formalized manner. It

serves as a bridge between the underlying ontology model and to the end-to-end users and enabling individuals with technical expertise in semantic technologies to perform, and express compliance checks and interprets the results. By presenting outcomes and the results in a clear and accessible format, the interface enhances the usability and practical applicability of the system making it more suitable simpler for the real-world use

The proposed and developed ontology-based approach is quite beneficial and it improves consistency and reduces ambiguity in the interpretation of legal rules and conditions. That approach leads it to automated and scalable compliance and, minimizing manual effort and supporting more potential and more reliable decision-making processes.

However, on the other hand there are also some certain limitations are present, particularly in capturing and focusing the full complexity and contextual interpretation of legal reasoning.

As this model is mostly rely and dependent on formal representation on formal representation, which may not fully reflect the nuances of real-world legal scenarios.

Overall, this research demonstrates that ontology-based approach, offer a practical and effective way to bridge the gap between legal text and computational systems. While certain limitations remain particularly in capturing the complexity of legal reasoning. This study provides a solid foundation for the future work in legal informatics and automated compliance solutions.

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# Glossary

| | |
|---|---|
| Agent | An entity that performs actions, such as engaging in credit activities |
| Class | A category or concept in an ontology representing group of entities |
| Data property | A property that links an individual to a data value |
| Individual | A specific instance of a class in ontology |
| License | A legal authorization to perform an activity |
| Object Property | Relationship between two individuals |
| OWL | A language used to create and define ontologies |
| RDF | Standard Model for representing data as triples |
| Reasoner | Tool to infer new knowledge based on logical rules |
| Semantic Web | A framework that enables data to be shared and understood by machines |
| SPARQL | A query language used to retrieve and manipulate RDF data |
| Taxonomy | A hierarchical classification of concepts or entities |
| Penalty | Sanction imposed due to violation of rules |

*Table 1: Glossary*

# Abbreviations

| | |
|---|---|
| OWL | Web Ontology Language |
| RDF | Resource Description Framework |
| SPARQL | SPARQL Protocol and RDF Query Language |
| LODE | Live OWL Documentation Environment |
| Widoco | Wizard for Documenting Ontologies |
| GUI | Graphical User Interface |
| DL | Description Language |
| NCCP ACT | Australian National Consumer Credit Protection Act |
| RDFS | RDF Schema |
| ASIC | Australian Securities and Investment Commission |
| ACL | Australian Credit License |

*Table 2: Abbreviati*

# Chapter 1: Introduction

## 1.1 Background to Legal Ontologies:

Systems are inherently complex, consisting of numerous statutes, regulations, and case law that govern human behavior. **Ontology** in terms of computer science and Information Science, is a formal representation of as a set of concepts within a domain and the relationship between those concepts. It Helps us structure and organize information to make it understandable by both humans and machines. When it comes to legal systems it becomes more complex and complicated, **Legal ontologies** are formal representations of legal concepts, relationships, and rules, designed to facilitate knowledge sharing, reasoning, and automated decision-making in the legal domain.

From last couple of years, the adoption of ontologies and data representation as set of concepts in Legal informatics has grown rapidly. Legal ontologies allow you to model obligations, permissions, prohibitions and exceptions such that reasoning engines can automatically classify entities or identify possible violations. As an example, the concept of whether a person or entity is authorized to participate in some credit action under some legislation can be expressed formally with classes properties and logical constraints in an ontology.

Compliance checking, formal representation of data and other legal knowledge management are the areas where the ontologies are quite beneficial. It is thus a way of minimizing human error, ensuring compliance as well as offering the transparency and auditability of decision-making frameworks. There are some also other terms that we will see in the next sections.

## 1.2 : Semantic Web and Legal Knowledge Representation

Semantic Web is another term which describes another concept in which the data can be represented in terms of a website which has good representation and well-defined meaning and that enables the machines to process common information intelligently and effectively core technologies that helps us in this manner for Web semantics, are **Resource Description Framework (RDF)**, **Web Ontology Language (OWL)**, and **SPARQL** query language. They provide a better framework to represent Information in a suitable way as discussed. Web ontology language is a formal language used to create and share ontologies on the web, OWL is part of the **Semantic Web** standards and builds on **RDF (Resource Description Framework)** and **RDFS (RDF Schema)**.

**Key Differences Between RDF and RDFS**

| Feature | RDF | RDFS |
|---|---|---|
| **Purpose** | Represents basic data (triples). | Adds schema (structure and meaning). |
| **Focus** | Data representation. | Data modeling and relationships. |
| **Expressiveness** | Limited. | More expressive (hierarchies, domains). |

*Tabel 3: Key Difference between RDF and RDFS*

In the Legal domain, Semantic web technologies emphasis and facilitate the information, data and the other entities that are interconnected. The legal concepts or entities are such as "credit activity"," license"," defense" can be described as classes with relationships and constraint. For instance, any agent is going to engage in a credit activity without a license can automatically be classified as non-compliant agent while those other has some licenses etc. are permitted.

Semantic technologies thus provide a foundation for **ontology-based compliance checking systems**, where reasoning engines can infer whether specific actions conform to legal requirements.

# 1.3. Motivations of the Study

Financial practices and especially the credit activities are quite under observation and regularized. Regulators keep them on a tight leash to make sure everyday consumers don't get taken advantage of by unfair practices. Section 29 of the **Australian National Consumer Credit Protection Act 2009** that is also the case study of our project is enlightens and shows the prohibition of engaging in credit activities and doing transaction with the valid license or the proper authorization from the legal authorities. If such activities are practiced then there are some consequences of this violation in terms of the civil penalty and the criminal penalty. Despite the deep observations and the clear crystal rules, there are multiple violations evidenced by the **ASIC case against Cigno Australia and BSF Solutions**, where unlicensed credit activities led to fines and regulatory enforcement.

My thesis and the project are motivated by the need to **automate compliance verification** for Section 29. Traditional compliance processes are manual, error-prone, and slow. By representing Section 29 as a formal ontology, it is possible to model complex legal rules, including offences, penalties, and defenses, and implement reasoning systems that automatically detect compliance violations.

# 1.4: Research Objectives:

The primary and the main objectives of this thesis are:

1. **Ontology Modeling:** The first main objective of this thesis is to model/design a formal OWL ontology Representing Section 29 of the Australian National Consumer Credit Protection Act 2009, including relevant concepts such as agents, credit activities, licenses, offences, penalties, and defenses.

2. **Compliance Checking:** The Data that is represented to develop a system capable of reasoning over the ontology to automatically identify complaint and non-complaint agents are compliance and working properly as expected
3. **Visualization and Documentation**. To provide clear, human-readable representations of the ontology using tools like yEd for diagrams and Widoco for documentation
4. **Case Study Validation:** To test the ontology and compliance checking system using real-world scenarios, including the ASIC enforcement case against Cigno Australia and BSF Solutions.
5. **UI Interface:** we will also develop a webpage that shows that how the data is represented and how to fetch and how to compliance the data in a proper way. The need for the UI interface is to just show how data is represented to a Non-It Guy and other persons for querying and displaying compliance results.

# 1.4: Structure of the Thesis:

This thesis is organized into the following chapters:

- **Chapter 2: Legal Background** – Provides an overview of the Australian National Consumer Credit Protection Act 2009, with a focus on Section 29, offences, penalties, and defenses.
- **Chapter 3: Related Work** – Reviews previous research in legal ontologies, ontology-based compliance checking, and Semantic Web applications in law.
- **Chapter 4: System Architecture** – Describes the components of the compliance checking system, including ontology, reasoning engine, triple store, and user interface.
- **Chapter 5: Ontology Design and Implementation** – Details the ontology development process, class hierarchy, properties, and reasoning logic.
- **Chapter 6: Ontology Visualization Using yEd** – Presents diagrams representing the ontology structure and relationships.
- **Chapter 7: Ontology Documentation Using Widoco** – Discusses metadata generation and automatic documentation.
- **Chapter 8: Compliance Checking Implementation** – Explains Graph DB triple store setup, SPARQL queries, and reasoning mechanisms.
- **Chapter 9: User Interface Implementation** – Describes the web interface for querying and displaying compliance results.
- **Chapter 10: Case Study: Section 29 Compliance Checking** – Demonstrates system validation through real-world scenarios, including licensed and unlicensed agents and legal defenses.
- **Chapter 11: Evaluation and Discussion** – Analyzes the effectiveness, advantages, and limitations of the system.
- **Chapter 12: Conclusion** – Summarizes contributions, findings, and potential future improvements.
- **Chapter 13: Bibliography and Useful Links** – Lists all references, including legislation, academic papers, tools, and web resources.

# Chapter 2: Legal Background

## 2.1: Overview of the Australian National Consumer Credit Protection Act 2009

The **Australian National Consumer Credit Protection Act 2009 (NCCP Act)** is one of the key legislative frameworks regulating consumer credit activities in Australia. It was an initiative ensure that credit holders and providers, brokers and other who are going to involve in a credit activity must practice a fair transparent and responsible manner while dealing with the consumers. The main objective to refrain the brokers from unfair means and other bad practices and also make the credit activities reliable and trustworthy.

Before the introduction of the NCCP Act, credit regulation in Australia was largely fragmented across different states and territories. This act was regularized and established a national level regulatory framework an observatory structure to formalize the rules and aspects handling the credit activities and transactions between different consumers and firms. Under this Act and regulatory framework, entities and other involving personalities must follow these rules and comply with the rules and requirements.

After this Act was proposed the administering and enforcing this framework the **Australian Securities and Investments Commission (ASIC)** was a responsible authority behind this framework's primary objective of the ASIC was to ensure that individuals and other organizations and entities who are going to engage in credit activities must comply and follow the rules and other obligations. Moreover, they have also the permissions to initialize the legal checkups and pursue legal proceedings that who are violating the Law by initiating enforcement actions, making huge fines and imposing sanctions and penalties.

A main feature of the Act that was required is the presence for businesses or individuals engaging in certain financial services to obtain an **Australian Credit License (ACL)**. This was an approach that only qualified and regulated authorities, the responsible and reliable entities are allowed to take part in the credit markets. There was also a framework so that who are aspirant for taking a license must follow some rules and regulations, including maintaining adequate competence, complying with responsible lending requirements, and acting honestly and efficiently and show professionalism when dealing with consumers.

Moreover, in addition to these licensing provisions, the Act also introduces mechanism and a structure to regulate the contracts that include credit activities, consumer leases, and other mentioned responsible lending practices. The provisions aimed at some constraints that reduce financial dodging and harm and ensures that the consuming activities must not exposed to unsuitable financial products and frameworks.

For the purposes of this thesis, particular attention is given to **Section 29 of the NCCP Act**, which establishes a prohibition against engaging in credit activities without holding a valid license.

## 2.2: Section 29: Engaging in Credit Activities Without a License

Section 29 of the NCCP Act establishes a fundamental rule within the Australian credit regulatory system: **a person must not engage in credit activities unless they hold a valid license authorizing them to do so**. This structure forms the main core logic of the legal norm that the ontology developed in this thesis aims to model and evaluate. The Simple and conceptual representation of the section 29 that engaging in a credit activity is a rule that can express as follows

*"IF PERSON ENGAGES IN CREDIT ACTIVITY AND PERSON DOES NOT HOLD A VALID LICENSE THEN VIOLATION OCCURS (OFFENCE"*

However, the law also recognizes exceptions that prevent the rule from applying in specific circumstances.

## 2.2: Legal Requirements for Credit Activities:

The NCCP act specifies and regulates that the individuals and organizations must have an Australian credit License from the respective concerned authorities if they want to engage in a credit activity. The Activities are of many types and can have many ways to indulge different services. Some of the examples of that services are providing credit to consumers, for their usage or may also act as intermediary between lenders and borrowers or offering services related to credit contracts and the constrains that are defined.

For the Licensing requirement it serves several important purposes:

1. **Consumer Protection:**

   Licensing ensures that that the individuals providing the services that include credit practices must follow the ethical and professional standards. The consumer protection practices through regulating transparency.

2. **Market Integrity:**

   By regulating participants in the credit market, the Act helps to reduce the risk of fraud, exploitation, and irresponsible lending and other vulnerabilities

3. **Regulatory Oversight:**

   The entities that are carrying license are subject to monitoring by ASIC and must comply with reporting and operational requirements and the constraints that are designed to formalize the compliance.

There are some specific mandatories or you can say the obligations for those entities who hold a license, they must have to comply with them in an appropriate way including maintaining internal procedures, ensuring the competence of the staff and providing the accurate information to the consumers. Failures to these obligations or in case of incompliance of these constraints leads to the revocation of the license or face some penalties. The penalties are of different types like civil penalty or criminal penalty.

In the context of the ontology modeling, the licensing requirement can be represented as a relationship between and Agent and License, where the in the absence of the that required license determines whether the Agent is permitted or not to perform the relevant credit activities.

## 2.4: Offences, Penalties and Defenses:

The Section 29 of the Australian National Consumer Credit Protection Act 2009 establishes both the Civil and criminal procedures in case of the violation of the licensing requirements that are required by the Act. This law recognize that someone want to engage in the credit activity without a proper justification or authorization can cause significant harms or debit to the consumers and affects the credibility and undermine the trust in the financial markets. The penalties that are discussed in the act are described below and we will a simple intro of them as we don’t want to dive deep in the act as our main focus in the ontology but to understand the process, we must know the pros and cons of the subject

- **Civil Penalty**

If a person engages without having a valid credit license do a credit activity that involves the financial means, then he must be subject to a civil penalty of up to 5000 penalty units. Civil penalties that occurred in violating the rules are mostly imposed through the regulatory enforcement proceedings that are initiated and governed by the ASIC.

- **Criminal Offence**

In additions to the civil penalties that are mention earlier above, the section 29 also establishes a criminal offence for the violators' person or individual commits a criminal offence if:

i. They are required to hold a license under Section 29
ii. They engage in conduct
iii. The conduct contravenes the licensing requirement.

The criminal penalty for this offence may include imprisonment for up to two years. There are some also legal defenses which are discussed below

- **Legal Defenses:**

The law recognizes and identifies some certain and clear situations where an individual may engage in a credit activity without holding personally a valid license. These Defenses ensured that the legitimate business arrangements are not unfairly paralyzed.

There are two main scenarios that are exists and described

**Defense 1: Acting on behalf of a licensed Principal**

A person may legally perform credit activities if:

- They act on behalf of another entity (the principal).
- They are an employee, director, or credit representative of the principal.
- Their actions fall within the authority granted by the principal.
- The principal holds a valid credit license.

**Defense 2: Acting for an exempt Principal**

A person may also be exempt if:

- They act on behalf of a principal.
- The principal has been granted a regulatory exemption under specific provisions of the Act.

I will draw a simple table for better understanding

| Sr# | Condition | Legal Outcome |
|---|---|---|
| 1 | Agent has License | Permitted Activity |
| 2 | Agent has no license | Potential Violation |
| 3 | Agent acts on behalf of licensed principal | Defense Applies |
| 4 | Agent acts for exempt principal | Defense applies |
| 5 | Agent has no license and no defense | Offence |

*Table 4. Logical conditions of the System*

# Chapter 3: Related Work

## 3.1 Legal Ontologies in Knowledge Representation

There is growing demand of the tools and ways to the solutions that are required that can represent the legal knowledge and formalities in in structured and well-mannered way that the can be represent in an understandable way and machine-readable format without increasing the complexity of the legal systems. The Most promising and enlighten approach is the use of the legal Ontologies which provide in a better way the formal models and structures for representing the legal concepts and the relationships and the rules among them.as we discussed that there multiple ways to describe the knowledge representation that how a data can be represented but here I show you with simple diagram from the journal “A Systematic Review of Knowledge Representation Techniques In Smart Agriculture nu Bikram Pratim bhuyan , Ravi thomar and amar ramdane cherif.

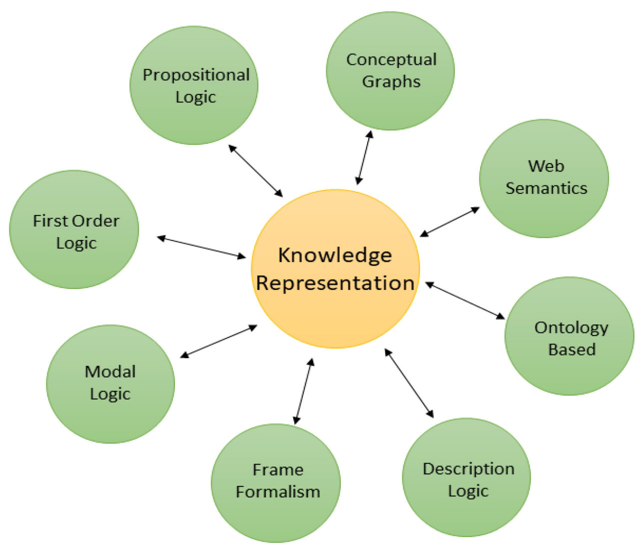


*Figure 1.*

From the above picture as we see will focus and the ontology-based and web semantics knowledge representation way.

In simple words if we say, the ontology can be defined or you can say described as a formal specification of shared conceptualization of a specific domain. Within the legal domain, the ontologies are roughly used to represent entities such as legal actors, obligations, permissions, prohibitions and institutional relationships and other related constraints. By Structuring and formalizing the legal knowledge and concepts in this way the ontologies enable computers and machines to interpret legal concepts and perform automated reasoning tasks.

There are plethora of areas and sectors where the legal ontologies have been applied and implemented, if we discuss some of them are legal information retrieval, regulatory compliance checking, contract-analysis and knowledge representation and knowledge management systems. One of the main and key advantages of ontology-based approaches is that they allow and permit the legal norms and laws to represented in a structure format so that it can be interpreted by reasoning engines. This enables the development of systems capable of identifying legal irrelevances and inconsistencies or detecting violations automatically in a quick and automated way.

From earlier discussions and picture shown above there also several research efforts that have explored how ontologies can represent legislative and regulatory structure and frameworks. These works are often relied on the semantic web technologies such as RDF and OWL to model and formalize the relationships between legal entities and actions. Through this way the legal knowledge can be integrated across systems and shared among different platforms and applications as required

Even there are the advantages of the modeling the legal norms and their compliance acts there are also number of challenges are present that may not easily explain and expressed using the simple and traditional knowledge representation techniques. As a result, researchers have proposed various modeling strategies to represent the structure and methodologies and meaning of the legal provision more efficiently and effectively.

The present thesis builds upon these developments by modelling a specific legal provision—Section 29 of the Australian National Consumer Credit Protection Act—within an ontology that supports automated compliance checking.

## 3.2 Ontology-Based Compliance Checking

The great and most difficult concern and challenge in many sectors is the regulatory compliance particularly in the financial services where the organizations and relevant institutions must adhere to complex legal requirements. usually, the compliance verification relies mostly you can say heavily on manual interpretation of the regulations by the experts especially the legal experts and compliance and regulatory officers. This process cannot be cost effective and it is time consuming and leads to the possibility of the human errors which it takes more time to cover.

An alternative approach or way method is offered and available by the ontology-based compliance checking by translating the legal requirements into formal logical structures that can be examined and processed by automatic process and automated systems. In such systems. Legal rules are represented as ontology classes, properties and logical restrictions. Reasoning engines can then evaluate whether a specific scenario satisfies or violates these conditions and constraints.

The main advantage of this approach is the possibility of **automated reasoning**. There is something that once the legal rules are represented within an ontology, reasoning tools can infer relationships and that are not explicitly stated, to understand this in a better way let me explain via an example that if a person engages in a credit activity without holding a valid license, the reasoning engine can automatically classify that person as non -compliant agent.

Ontology-based compliance checking systems have been applied in several domains including financial regulation, environmental compliance, and data protection laws. These systems typically integrate three key components:

1. A **legal ontology** representing the regulatory framework
2. A **reasoning engine** capable of evaluating logical conditions
3. A **query interface** that allows users to test compliance scenarios

The approach used in this thesis follows this general structure. The legal rule contained in Section 29 is represented in an OWL ontology, while compliance checking is performed using reasoning mechanisms and SPARQL queries. We will see later more about in the upcoming sections.

## 3.3 Semantic Web Technologies in Legal Applications

In this section we will discuss the semantic web technologies in legal applications and how there is close relationship or closely linked between the ontology-based systems and the emergence of the semantic web technologies. The Semantic Web is an extension of the traditional web in which data is structured in a way that allows machines to understand relationships between different pieces of information.

Several technologies play a central role in Semantic Web applications:

**RDF (Resource Description Framework)**
RDF provides a data model based on triples consisting of subject, predicate, and object. This structure allows information to be represented as interconnected data. RDF is a framework for representing information on the Web in a structured and machine-readable way. It is a fundamental technology of the **Semantic Web**, allowing data to be shared and linked across different systems and domains

**Key Concepts of RDF**

1. DATA REPRESENTAION
   a. Triples
      i. Subject
      ii. Predicate
      iii. Object
2. URI INFORMATION (Uniform Resource Identifier)
3. Graph-Based Structure
4. Syntax Format
   a. RDF/XML
   b. Turtle
   c. JSON-D

**What is RDF Schema (RDFS=** Resource Description framework schema)

RDF Schema (RDFS) is an extension of the Resource Description Framework (RDF) that provides a basic vocabulary for describing the structure, meaning, and relationships of RDF data. It enables the creation of simple ontologies by defining classes, properties, and hierarchies.

**OWL (Web Ontology Language)**
OWL is a formal ontology language that enables the definition of classes, properties, and logical constraints. OWL also supports reasoning mechanisms that allow systems to infer new knowledge from existing data.

**SPARQL**
SPARQL is a query language used to retrieve and manipulate data stored in RDF format. It allows users to search for patterns within semantic datasets.

These technologies are particularly useful in the legal domain because laws often involve complex relationships between actors, actions, and conditions. Semantic Web frameworks allow these relationships to be represented explicitly and evaluated through automated reasoning.

In the context of this thesis, OWL is used to model the legal ontology, while SPARQL queries are used to evaluate compliance scenarios stored in a triple store.

**Semantic Web Languages:**

**How These Languages Work Together**

1. **RDF:** Basic data representation**.**
2. **RDFS:** Adds meaning (schema) to RDF data**.**
3. **OWL:** Enables advanced reasoning and complex relationships.
4. **SPARQL:** Allows querying and extracting data.
5. **SHACL:** Validates the structure of RDF data**.**

**Provisions:**

A provision is a specific class or requirement in a legal document or contract that outline that must be done or what is allowed

**Example:** A provision in a rental agreement that the tenant must pay rent by the 5th of each month.

One more thing that should must be understand that RDF is intended for situations in which information on the Web needs to be processed by applications, rather than being only displayed to people.

## 3.4 The Compliance Checking Pattern by Francesconi and Governatori

There is a particularly influential and quite great beneficial contribution to ontology-based legal compliance modelling is the work done by the Prof.Enrico Francesconi and Guido Governatori, who proposed a pattern for pattern for representing legal compliance rules within a decidable Semantic Web framework.

The main focus of both of them is on how to focus on how legal norms and ethics can be translated into logical structures that remain compatible with OWL reasoning mechanism and structures. One of the key challenges in modelling legal norms is that laws often contain complexities and defeasible rules, that are hard to understand and even formalized, meaning that general rules may be over bidden by specific exceptions or defenses.

The compliance checking pattern proposed by Francesconi and Governatori addresses this challenge by using class OWL restrictions and constraints that are representing legal norms and entities. Instead of encoding rules as procedural logic, the approach model's compliance conditions as ontology classes whose membership is determined through reasoning.

In this pattern, legal entities are represented as individuals belonging to different classes depending on their legal status. For example, an individual may be classified as:

- A **permitted agent** if all required legal conditions are satisfied
- A **non-compliant agent** if the conditions are violated
- An **offending agent** if the violation corresponds to a legal offence

Reasoning engines can then infer which class an individual belongs to by evaluating the logical conditions defined in the ontology.

This is a very good approach and methodology that is suitable for the legal entities and the compliance related to these entities because it allows the system to automatically determine whether a given scenario satisfies the legal requirements

Furthermore, the approach ensures that reasoning remains within the **OWL 2 DL** framework, which guarantees computational decidability and integrity regarding the results.

The ontology developed in this thesis follows this compliance checking pattern. The legal rule defined in Section 29 is represented using OWL class restrictions that distinguish between compliant and non-compliant agents based on whether they hold a valid license or fall under a legal defense. We will see the physical implementations later.

## 3.5 Challenges in Modelling Legal Rules

There are multiple advantages and benefits of ontology-based approaches but modelling legal rules within formal ontologies is a challenging task, as it requires a lot of understanding and it has a lot of complexities. there is ambiguous language, the difficult words that are not even easy to pronounce or understand by a common man as they include complex language and conditional structures and multiple layers of exceptions. Translating these elements into precise logical representations requires careful interpretation.

Along with the above factors under consideration there are is also another challenge arise with a fact that the legal reasoning frequently depends on contextual information that may not easy captured in formal models. To understand it in a simple way, let determine or consider that whether a person acted within the authority of a principal may require additional evidence beyond the information stored in the ontology. That is quite challenging.

Although in the presence of these limitations and constraints, ontology-based compliance systems remain a valuable tool for assisting regulatory analysis and ontology required framework. By formalizing or structuring the key or required legal conditions that are required, such systems can provide primary compliance assessments and help identify potential violations more efficiently.

## 3.6 Summary

This chapter reviewed existing research related to legal ontologies, Semantic Web technologies, and ontology-based compliance checking. Particular attention was given to the compliance checking pattern developed by Francesconi and Governatori which provides a structured method for representing legal norms within an OWL-based ontology.

The concepts discussed in this chapter form the theoretical foundation for the system proposed in this thesis. The following chapter introduces the architecture of the compliance checking system and describes the technologies used to implement the ontology and reasoning framework.

# Chapter 4: System Architecture:

## 4.1 Overview of the Compliance Checking System

The main aim of the thesis to formalize or demonstrate how legal rules and constraints can be transformed into a machine-readable format and used to automatically evaluate regulatory compliance. The architectures integrate semantic web technologies, ontology modelling tools and a simple web-based interface to create a prototype compliance checking system

In a high level advance specifications, the system operates in a manner in a way that it represents legal rules and constraints within an ontology and allowing a reasoning engine a disciplinary mechanism that is necessary to analyze or evaluate whether specific conditions or multiple scenarios satisfy the legal constraints and legal rules that are defined in the ontology.in our project thesis the core legal rule represented in this system is derived from the **Section 29 of the Australian National Consumer Credit Protection Act 2009**, which prohibits engaging in credit activities without a valid license and impose penalties in case of the violations. These violations are may be of criminal or civil.

There is a specific layered structure in the system defined in which each component has specific role and performs a specific role. This quite helpful in analyzing the legal rules that are described, also the ontology layer represents the legal knowledge, The data layer stores instances of legal scenarios, the reasoning layer evaluates compliance conditions, and the user interface layer allows users to interact with the system. We will also develop a UI interface in our thesis project just to show how it look to a user and the how to interact with it. I will also make it public by uploading it the github.com so that it is publicly accessible.

If we summarize the general workflow mechanism of the system it is described as follows to understand

i. Legal concepts and rules are modelled within an ontology using OWL.
ii. The ontology is stored in a triple store that supports Semantic Web reasoning.
iii. Users provide scenario information through a web-based form.
iv. SPARQL queries retrieve and evaluate relevant information from the knowledge base.
v. The reasoning engine classifies the scenario as compliant or non-compliant.

This architecture demonstrates how legal knowledge representation can support automated compliance verification in a structured and transparent manner. In the next section we will discuss some of the system components to easily understand.

## 4.2 System Components

The compliance checking system consists of several interconnected components that collectively enable the modelling, storage, reasoning, and evaluation of legal rules. Each component contributes to a different stage of the system workflow.

- **Ontology Layer**

  The Ontology layer forms the core of the system. This is the core part that contains the formalized and structural representation of legal concepts and rules that are derived from the section 29 of the NCCP Act. The Ontology Defines classes such as Agents, Credit Activity, License offence and Defense. It is not only the classes there are also relationships among them that are defined. In the protégé section I will describe these classes and we will also describe the relations that interconnect them.

  There also some other restrictions called logical restrictions that are defined within the ontology to represent the legal conditions under which an Agent is considered compliant or non-compliant. For better understanding let's describe that the agent who is engages in a credit activity without holding a valid license may be classified as violating the legal rule.

- **Data Layer**

  The data layer is another layer that stores instances representation specific and formulated legal scenarios and conditions. These instances may describe individuals, some specific identities and organizations, firms performing, certain actions within the credit market and in the financial market.

  For example, an individual or simply a singular entity may be represented as performing or engaging in a credit financial activity while either possessing or lacking of a valid authorized license. The reasoning engine can then evaluate these instances against the rules defined in the ontology. In the upcoming section let's have some discussion about the reasoning layer.

- **Reasoning Layer**

  The Reasoning layer is another layer that evaluates logical relationships between the concerning entities inside the ontology. Using an Owl reasoner, the system can automatically infer class memberships based on the restrictions defined in the ontology.

To understand this in a better way lets go through a simpler way that if an individual engages in a credit activity and does not have a valid license, the reasoner will infer that the agent belongs to the non-compliant class. Similarly, if an agent acts on behalf of a licensed Principal, the reasoner may classify the agent as permitted, we will practice those reasoners in the upcoming sections in the protégé, and the names of the reasoners that are used are pellet and Hermit.

- **Query Layer**

This layer in the ontology is used to allows the users to retrieve information from the knowledge base. SPARQL queries are used to search for patterns and combinations that are inside the stored data sets and extract relevant compliance results.

These queries enable the system to identify whether a given scenario satisfies the conditions required for lawful participation in credit activities.

- **User Interface Layer**

The final layer of the system is the user interface, which provides a simple mechanism for interacting with the compliance checking system. Through a web-based form, users can input information describing a particular scenario.

The system processes this information and returns a compliance result indicating whether the agent's actions conform to the legal requirements defined in the ontology. we also show a website where we can find that type of User interface Results in the web, we will develop two interfaces one with the drop-down buttons in which we make some Boolean conditions to show our ontology in a better way so that the non-programmers can understand it easily and also, we will make a user input form so that we can add the data or in the form of a new user who can add the inputs and then can have the required results by inputting, the data.

Now we will discuss the technology used in our project that technologies that we were used and going to use.

## 4.3: Technology Stack:

The system integrates several technologies that support ontology development, knowledge storage, reasoning, and user interaction. Each tool contributes to a specific part of the overall architecture.

## 4.3.1: Tools Used

The following tools were used during the development of the compliance checking system.

### 4.3.1.1: Protégé (Ontology Development)

Protégé is an open-source ontology editor widely used for developing knowledge models in the Semantic Web domain. It provides a graphical environment where classes, properties, and logical restrictions can be defined.

In this thesis, Protégé was used to construct the ontology representing legal rules derived from Section 29 of the NCCP Act. Classes such as Agent, License, and CreditActivity were defined, along with object properties describing relationships between these entities.

Protégé also supports reasoning plugins that allow developers to test ontology logic and verify whether individuals are correctly classified according to defined rules.

Let's have some, the visual representation of the classes and other relationships among them and about the object properties, data properties and individuals', these are the real-time classes and individuals and object properties that are used in the project thesis I will have some screen shorts of them for easily understand.

There is also a term that called rdf resources which means that all the things that are described by the rdf are called resources and are the instances of the class rdf: resource.

- **Classes:** Resources may be divided into groups called classes. The members of a class are known as *instances* of the class. Classes are themselves resources. They are often identified by IRIs and may be described using RDF properties. The rdf:type property may be used to state that a resource is an instance of a class.

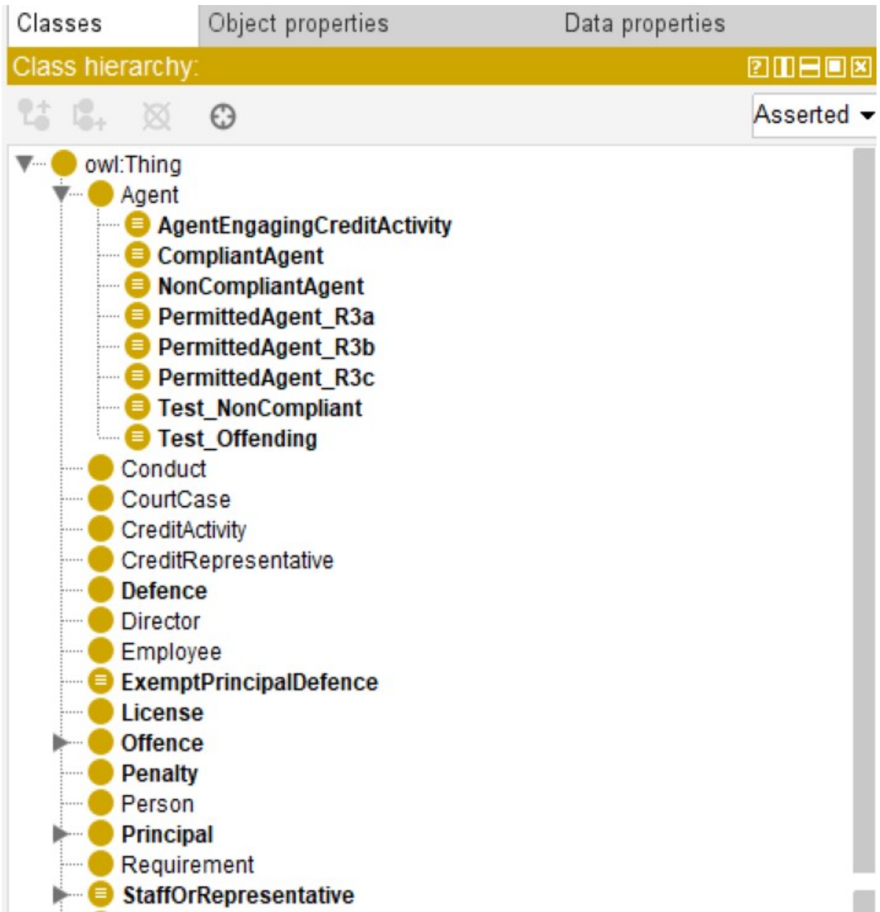


*Figure 2: Class hierarchy*

## ➢ Data Properties:

In the rdf section there are handlers like Literals and datatypes, The main purpose is that they connect an individual to a specific value or a piece of raw data rather than the other indivuduals. here is the pictorial glimpse of the data that we used in our ontology

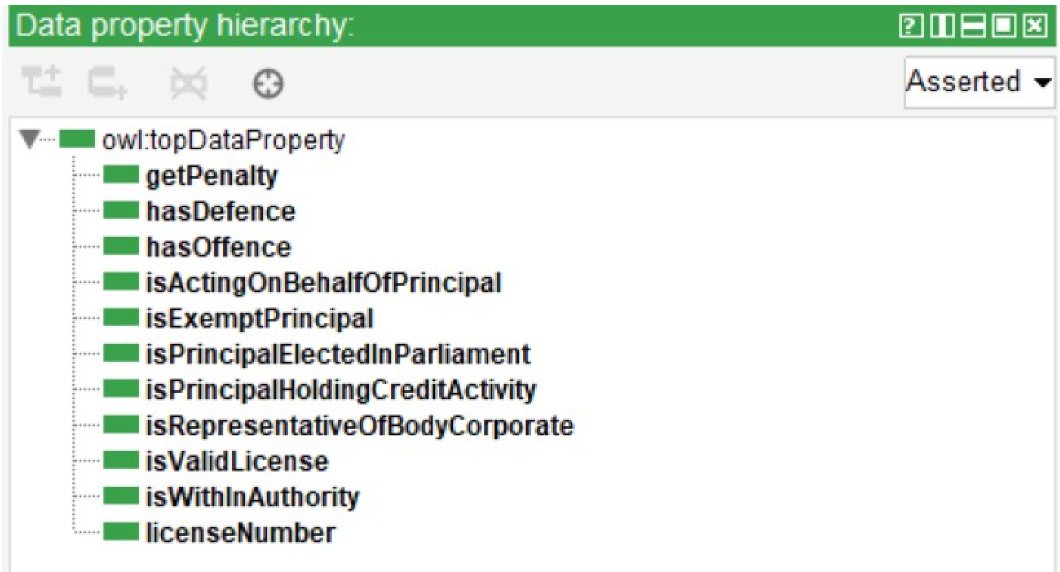


*Figure 3 Data Properties Hierarchy*

## ➢ Object Properties:

The Text refers to these simply as properties that relate one resource to another. In simple words they act like a simple bridge between two individuals they describe a relationship between two things. Here attached is also a pictorial glimpse of our object property that we used in our ontology building


*Figure 4: object properties hierarchy*

➢ **Individuals:**

The individuals are the actuals things object in our data we can say that these are the real acting properties that are going to act to check the ontology. They everything that you talked about is a specific resource that talk about in an individual's course

## 4.3.1.2 OWL 2 DL

The ontology developed in this project is expressed using **OWL 2 DL**, a formal ontology language designed for representing complex knowledge structures while maintaining computational decidability.

OWL 2 DL allows the definition of logical structural condition constraints, class hierarchies, and relationship between the properties that enable automated reasoning in the ontology. This is amazing as it sometimes the system infers new knowledge from existing factors stored in the ontology

In the context of this thesis, OWL restrictions are used to represent legal norms such as licensing requirements and defense conditions.

## 4.3.1.3 Graph DB

Graph DB is a semantic graph database designed to store and manage RDF data. It acts as the triple store where the ontology and its instances are stored.

It provides efficient indexing mechanism and supports reasoning over different and interconnected ontologies also offer the integrated SPARQL queries endpoint that allows users to query the stored data and infer results.it is run on the localhost and we will also attach the graph DB web page. the process is that we save the ontology in the form of Owl and it can export to the Graph DB and we can infer results from the data representations

Within this system, Graph DB is responsible for managing the knowledge base and executing SPARQL queries used in compliance checking. we will discuss the SPARQL queries in our next section.

## 4.3.1.3 SPARQL QUERIES

SPARQL is a query language used for retrieving and manipulating data stored in RDF format. It enables users to search for patterns in semantic datasets.

In this thesis, SPARQL queries are used to retrieve information about agents, licenses, and credit activities stored in the ontology. These queries help determine whether a particular scenario satisfies the conditions required for legal compliance. SPARQL "hunts" for data using the terms we talked about earlier (**Individuals**, **Object Properties**, and **Data Properties**).

### 4.3.1.3 yEd Diagram (Ontology Visualization)

yEd is a diagramming tool used to create conceptual representations of ontology structures. It allows developers to visually represent classes, relationships, and hierarchical structures within the ontology.

In this project, yEd was used to create diagrams illustrating the relationships between legal entities such as agents, licenses, credit activities, and offences.

These visual representations help clarify the conceptual structure of the ontology

### 4.3.1.6 Widoco (Ontology Documentation)

Widoco is a documentation generation tool designed specifically for ontologies. It automatically produces human-readable documentation describing ontology classes, properties, and relationships.

Using Widoco, the ontology developed in this thesis was documented in a structured format that allows readers to explore the ontology structure and understand its components.

### 4.3.1.7 HTML, CSS, and JavaScript (User Interface)

As we discussed earlier, we will develop a simple interface for our ontology representation A simple web-based interface was developed using HTML, CSS, and JavaScript. This interface allows users to input scenario information and submit queries to the compliance checking system.

The interface interacts with the Graph DB SPARQL endpoint and displays the results returned by the query engine.

### 4.3.1.8 GitHub

GitHub was used to manage the source code and ontology files developed during the project. It provides version control and allows the project repository to be shared and maintained efficiently.

GitHub also enables automated workflows that can generate ontology documentation using Widoco.

## 4.4 Chapter Summary

If we summarize the whole discussed above chapter, the chapter involves the structural and modeling architecture of the ontology-based Compliance checking system that we developed in the system, we used number of technologies that we used in our system. about these technologies we gave a number of reasons and discussions above in the report. The system integrates several components including an ontology model, a semantic triple store, the reasoning mechanism and a user interface.

By Combining these and interconnect these technologies and tools the system demonstrate how legal rules can be represented formally and evaluated automatically. The next chapter focuses on the design and implementation of the ontology itself, describing the classes, properties, and logical restrictions used to represent the legal provisions of Section 29.

# Chapter 5: Ontology Design and Implementation

This chapter demonstrate how the ontology is designed and implemented.

## 5.1 Ontology Development and Methodology:

The development of the ontology in this thesis followed a structured and formalized methodology commonly used in Semantic Web projects to represent the data and the knowledge. The main objectives were to translate the legal rules defined in the section 29 of the Australian National Consumer Credit Protection Act 2009 into a formal representation that can be interpreted by reasoning systems.

There are several steps that are needed to develop an ontology that is fully integrated with the system. First the relevant legal concepts and the relevant legal entities are carefully identifying through careful analysis of the data given or the given legislative text. These concepts included entities such as agents, credit activities and offences and legal defenses. Each of these concepts was then represented as a class within the ontology.

Furthermore, the second step involved defining relationships between those concepts and technologies. Object properties are used to express these relations that how different object or different entities interact with each other with inside the legal framework. For example. an agent may hold a license or engage in credit activity.

These relationships show how the entities and relationships are interconnected with each other and once this basic conceptual structure is established, logical restrictions were introduced to represent legal conditions. These restrictions allow the systems and the reasoning engines to infer whether an Agent is Compliant or non-compliant with the legal rules.

The ontology was implemented using **Protégé**, which provides a graphical environment for creating OWL ontologies. Protégé also allows developers to test reasoning outcomes using built-in reasoning engines.

The development methodology adopted in this thesis can be summarized as follows:

- **i.** Identification of legal concepts from the legislation
- **ii.** Creation of ontology classes representing legal entities
- **iii.** Definition of object and data properties describing relationships

**iv.** Introduction of logical restrictions to represent legal rules
**v.** Testing and validation of reasoning results using Protégé

This structured approach ensures that the ontology accurately reflects the legal requirements defined in Section 29.

## 5.2 Ontology Classes

The ontology contains several classes as we already discussed in the previous chapter and these classes represents the main legal concepts involved in the regulation of credit activities. These classes will define the legal entities and elements that are going to participate in the legal framework and it also serves as the main foundation for the modeling compliance conditions and constraints that are defined in the ontology by the given text.

One of the main and the central class in our ontology is the Agent Class, this class represents individuals or organizations that may engage in financial credit activities. Agents are the primary actors whose behavior must be evaluated and examined for compliance with the licensing and other requirements.

Another one of the most important and crucial class is the creditActivity which represents that the Actions falls within the criteria or scope of the NCCP act. These activities may include providing credit services, facilitating credit contracts or acting as an intermediary in financial transactions. This is also the main bridge class between the relations between the entities and relationships.

Our ontology also includes a license class representing the legal authorization required to engage or involved in financial or regulated credit activities. An Agent that holds a valid license may legally perform are act legally or in the regularities of the NCCP act. The Agent may perform certain actions with in the market

Additional classes are used to represent legal consequences and exceptions defined by the law. These include:

- **Offence**, representing violations of the legal rule
- **Defense**, representing legal justifications that prevent liability
- **Principal**, representing the entity on whose behalf an agent may act
- **BodyCorporate**, representing corporate entities that may hold licenses

To support compliance reasoning, the ontology also defines specialized subclasses that represent different legal outcomes. For example, an agent may be classified as compliant if they hold a valid license or fall under a legal defense.

The hierarchical structure of the ontology can be illustrated as follows:

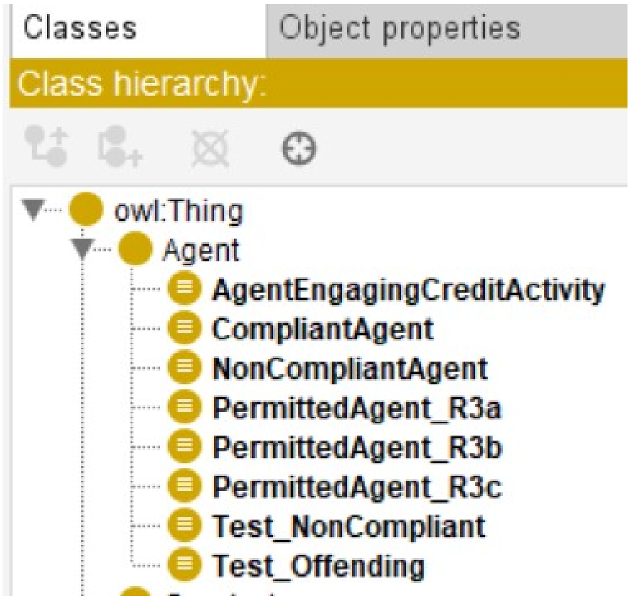


*Figure 5: Main classes related to Agent*

These subclasses represent different scenarios in which an agent is permitted to engage in credit activities. we will also see where the classes and other individuals that we are already defined in the ontology infer in the concerned sub class.

## 5.3: Object Properties

Object properties define relationships between the individuals of the ontology that belong to different classes. In the Ontology that we developed for our thesis, and we define classes and object properties, these object properties play an important and crucial role in representing and modelling and formalizing the interactions and relationships between the legal entities, legal actors and the regulatory requirements

To define the relationships in the section 29 multiple objects are defined in the ontology. One of the main and crucial object properties is the acts on the behalf of the someone, it means that Agent may act on behalf of someone else who hold a license or who have a valid license to engage in the credit activity. As we already studied in the act that an agent can be applicable to engage in a credit activity on behalf of some on and can take participate in the financial market and the financial credit activities. The person or acts on someone behalf is may be the principal or director but it does not give the person that it has authority to participate.

Another property is the has License, it is a property that links that Agent to the license which authorize it to their credit activity.

Moreover, the property has Engage in represents the relationship between an Agent and the Credit activity they perform. This is a property of different type that shows that it has engaged in a credit activity, as it allows the system to identify whether an Agent has participated in actions that fall under the scope of the legislative criteria and the constraints.

Other object properties used in the ontology include:

- **hasOffence** – links an agent to an offence when a violation occurs
- **hasPenalty** – connects offences to their corresponding penalties
- **hasDefense** – indicates that a defense applies to an agent
- **committedBy** – links an offence to the agent responsible for the violation
- **involves** – connects legal events with the entities involved

These properties allow the ontology to represent complex legal relationships in a structured manner.

## 5.4 Data Properties

Data properties are used to associate individuals with literals values such as numbers text or Boolean expressions, in simple words the data properties provide additional information that supports the reasoning and classification in the ontology for proper inferring.

Here is simple glimpse of our data properties that we used in our ontology.

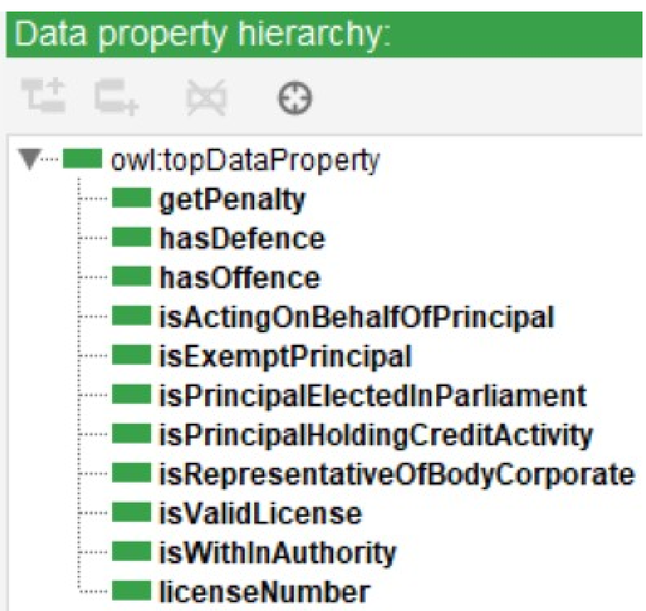


*Figure 6: Data Properties showing relationships*

For example, the property isValidLicense indicates whether a particular license is valid under specific conditions. This1 entity or property may represented as a Boolean value that determines whether the license satisfies the regulatory and other requirements that are necessary for the modelling.

The other property is the getPenalty which means that the if some violates the laws that are inferred properly in the ontology, they must get penalty and this data property is also used to pursue according to the NCCP act, the penalty is of different types it is may be of civil penalty or criminal penalty.

Another property that is also quite important which is licenseNumber, this is important because it stores identifying information that is linked to a license, this property quite effective for the system as it allows to distinguish between multiple license numbers with the knowledge base or in the stored data so that infer properly.

This ontology also includes properties that describes characteristics of principles or corporate entities such as whether an entity is recognized as a representative of a cooperative body this is also very quite helpful because it helps to understand the ontology that who is a valid user or who is ordering a financial or credit activity on behalf of financial active license or representative that's why we use this in our ontology now let's jump to our next one.

As we describe above there are multiple ontologies and multiple data properties that we used in our ontology but there are few that for example the has defense and has offence are the two properties that can a user can entity can use in their ontology when involving in a financial credit activity the health defense credit activity or the data property is used when someone has the defense to do a financial or corrected activity and has offense if someone do without the credit activity he must have some offense or defense due to this activity and it will be a penalty for it.

## 5.5 Logical restrictions and Reasoning

As we discussed the multiple important and crucial features of the ontology and the use of ontology but the one of the most important features of the ontology is the use of logical restrictions and constraints to represent the legal rules that are discussed in the NCCP Act These frictions define and analyze the conditions under which an adjunct is considered compliant with the law.

For example, an agent or a legal entity maybe classified or determined as a permitted agent or agent who has some permit and if they involve or engage in a credit activity in the financial market and possess a valid license this condition can be expressed or verified using an OWL a class restriction that requires the presence of the relationship, they internal relationship between the agent and a valid license.

Similarly on the other hand defense conditions can be represented through restrictions and constraints involving relationships between the agents and principal in simple way fan agent acts on a beer for licensed principle and operates within the authority granted by that principle This system may classify the adjoint as compliant or the agent is valid for the credit activity.

Reasoning engines and systems evaluates analyze these complex logical conditions and automatically classify individuals according to the rules and regulations defined in the ontology. This allows the systems to infer automatically the legal outcomes without requiring minor evaluation of each scenario that saves times and cost effective.

The reasoning process is one of the most important particularly useful process for identifying a non-compliant behavior For example if an agent engages in credibility but all not foldable sense does not satisfy any defense conditions the system may classify the adjoint as violating the regal rules and laws and violations and it will be as a non-compliant agent we will use two reasoners in our project one is called hermit and second is called her pellet and these reasoners are used to infer properly the data automatically from the storage stored knowledge data set.

## 5.6: Ontology Implementation in Protege

As we discussed earlier in the technology stack section of our thesis about the Protégé the tool that was we are going to use for our ontology editor. The ontology developed in this thesis was implemented and formalized using the protégé ontology editor. Protégé is an open-source free software to edit ontologies and used for the representation of data as an on-ontology basis.

This software provides an initiative graphical that allows the users the developers to create classes define properties and define the data properties and individuals and specified logical restrictions and define the relationships between them which interconnects them.

During the implementation process, the classes and properties described in the previous sections were defined within Protégé. Logical restrictions were added to represent the conditions derived from Section 29.

After a formalizing and modeling the ontology structure the dust in visuals which we described in the ontology in the individual section work created to evaluate the reasoning behavior of the ontology. These individuals represent different compliance scenarios such as agents holding Like agents acting on behalf of Princip and agents engaging in unauthorized credit activities adjourned with some sort of specific offenses and adjuncts or entities with some sort of special defenses ontology.

After formalizing the ontology then the Protégé reasoning engine the Protégé reasoning system was then used to verify whether these individuals or entities were collectively correctly classified according to the logical rules defined in the ontology and in the description. We will also use some of our new individuals that belong to over the case study and also over ontology and will check that or they are inferring properly and giving the required results.

The results of these reasonings from this ontology tests confirm that the ontology successfully distinguish between the compliant and the non-compliant between the right result and between the result that is compromised and can be wrongly inferred.

## 5.7 Summary

This Chapter presented the design and implementation of the ontology developed for this thesis The ultimate goal was create that ontology to formally present or formalize the legal rules and the requirements defined in the section 29 of National Consumer Credit Protection Act 2009 and to enable some specific automatic compliance checking using web technologies.

The earlier in this chapter the ontology and development methodology were described which involved multiple relevant legal methodologies and concepts from the legislations and the main idea was to translating them into a structured and model ontology. The core legal entities such as corrected activities licenses offenses defenses and principal were modeled as ontology classes. There was also relationship between those entities using object properties and while additional descriptive attributes were modeled through data properties and other there were also individuals that were described to interconnect these entities to show their relationship.

Logical restrictions were also introduced to capture the legal models and legal conditions under which a specific user or you can say a specific agent can legally involve or engage in a credit activity in the financial market. These Const or restrictions allow the reasoning systems and reasoning engines automatically analyze the knowledge base data sets and infer whether an agent is compliant or not with the regulatory requirements.

So, in a nutshell if we conclude this the ontology was implemented using Protege ontology editor in which the classes properties individuals and logical constraints were formally defined in OWL. And the overall ontology design and implementation provide a structured and modeled a machine-readable representation of the legal rules that defined in this section 29 that demonstrates that the ontology is capable of distinguish between the legal requirements and forming the foundation automatically legal compliance checking in the system.

# Chapter 6: Ontology Visualization of yEd

In this chapter we will discuss about the visual representation the visual interface or you can see the ontology visualization through a program called yEd. Ontology representation or visualization to a user plays an important role in understanding or defining the conceptual structure of an ontology of a concept and relationships especially between different entities and between different objects within this system in the ontology.

While ontology editors such as Protege allow developers to create manage ontology entities and components, the graphical representation of an ontology helps provide a clearer and bigger initiative understanding of the knowledge model that is described in ontology from the Knowledge Base data set.

And the thesis the ontology developed for representing the legal requirements or legal entities defined in the discussed section 29 of the National Consumer Credit Protection Act 2009 is visually represented using the Y ed graph editor. The visualization and the representation allow the ontology structure to be represented in a more formalized a more formal and understandable and structured form. By representing the ontology graphically, readers can easily observe the hierarchy of the classes the relationship between different legal entities, and the overall structure of the knowledge model. Sometimes it is very difficult to analyze or to formalize a legal requirement into ontology but even for a common man or some person who is a non-developer the visual representation of a concept or of a process makes things easy to process to learn and to understand the workflow and how the work is done and how the data is represented and how the data is infringed from the required data sets.

The use of the conceptual diagram has many benefits but one of another benefit that is it also improves the readability of the ontology and helps illustrate and demonstrate how the different components interact with each other in order to represent the legal requirements and rules defined in the legal solution as we discussed earlier. We will draw the diagram and we will also discuss some points of the yEd diagram later in this chapter.

## 6.1 Purpose of Conceptual Diagram

Here a very beautiful question arises and that and what is the purpose of a conceptual diagram. As we discussed earlier in this chapter and the purpose of creating a conceptual diagram is to provide a visual overview of the ontology Structured Developed in our thesis. Even we discussed earlier and it is quite big fact that the legal ontology often contains several classes, properties and logical relationship that sometimes it is quite difficult you want to understand and when it is more difficult to analyze and understand when presented or formalized to form definitions.

Visual representation easier to understand for anyone and for a conceptual diagram and for our ontology in the YED helps us to illustrate the hierarchical structure of the ontology and the relationship between different entities even in the presence of some legal restrictions and some constraints involved in that legal framework. By visualizing the ontology it's become more easier to understand how legal concepts and legal rules expected from the legislations I'll translate it into formal ontology components.

Another one of the more important objectives of the conceptual diagram is to support explanation and to make confidence in the reasoning process of the ontology. Since the ontology includes logical rules and logical instructions that determine and analyze whether an individual or an agent is compliant or noncompliant with legal requirements, And the diagram helps demonstrate and illustrate how these elements are connected within the system.

The conceptual diagram therefore serves as an important bridge between the legal text and the formal ontology model implemented in Protege We will discuss more about this in our next sections.

## 6.2: Ontology Class Representation

In In this section, we will discuss the ontology visualization representing the main classes that were defined and formalized during the ontology design phase. Classes response to the primary the legal concept and legal rules and entities identified from the NCCP Act.

One of the central and one of the main classes in the ontology is the Agent class. This class represents individuals or organizations or persons that may engage in credit activities within the financial market. Agents or the individual or the organizations et cetera are the primary actors evaluated by the legal framework by the legal structure to determine whether they are legally authorized to perform regulated activities that are expected from them.

Going deeper one of the most important classes represented in the ontology is credit activity. This class represents financial actions and the activities that they fall within the regulatory scope or area of the Act. These activities our acts may include providing credited services Facilitating credit agreements or assisting consumers in obtaining credit.

The ontology also includes another very important and very crucial entity called license class that represents the legal authorization required for the agents to participate in the regulated credit activities. A valid license is mandatory or obligatory to perform a credit activity in the financial market those who hold a valid license may legally engage in credit related activities and whereas agents who perform these activities without proper authorization without proper license or without proper defense may violate legal provisions of the acts and they will face the legal penalties in terms of the civil and criminal penalty or both.

Moreover, there are some more additional classes in the ontology include Offense Defense, Principle and Body Corporate these classes represent legal consequences and exceptions organizational entities and other logical formal and model constraints and restrictions involved in the legal framework.

To support automated reasoning the ontology also includes subclasses that represents different compliance scenarios and that can also describe our ontology in simpler way for understanding and for explanation of the act in a better way. As the more explanation is an important factor of the ontology that the information is presented in small short steps so it can be easily inferred and retrieved and it can easily be represented in the knowledge base data sets. These classes represent legal consequences involved in the legal framework and exceptions as well.

## 6.3 Relationship Between Legal Concepts:

Moving forward in addition to representing legal entities two classes the ontology also defines relationships between these entities. These relationships are represented using object properties that connect individuals belonging to different classes.

For example, the property has less attribute that define the relationship between agent and Allison's that it authorizes the adjunct to perform the credit activity in the financial market even we earlier described these terms in the section of the object properties and data properties earlier in this this is report in the above chapters but to perform and to visualize the beauty and to demonstrate the Z diagram we will also discuss these here. The relationship is important because the presence of a valid license determines whether the agent satisfies one of the legal requirements for compliance.

Another important relationship is represented by the property acts on behalf of which describes scenarios and which describes situations where an entity or an agent performs activities on behalf of some specific principle. In such cases this complex terms the agent may still be completed compliant under certain legal conditions defined by the legislation but it shall not must be considered as compliant agent without these conditions beat your

The ontology also includes the property has engaged in which means which represents the relationship between agent and the credit activity and they are going to perform. The relationship helps determine whether the actions of an agent fall within the regulatory scope of the legislation.

There are also some relations such as offense, has penalty and has that also represents legal consequences and legal justifications that may affect the representation or inferring status or may affect the compliant status of an agent. These relationships allow the ontology to represent complex legal interactions in a structured and machine-readable format.

## 6.4 Explanation of the Ontology Graph

The ontology and graph created using yEd provides structured and modeled representation and formal visualization of the ontology developed in this thesis. The graph illustrates how the different classes and relationships are interconnected to present the legal undefined in the legislation. At the center of the graph there is a class Agent which is the main and access the primary entity with in to ontology from this central class multiple relationships connect the agent other legal entities such as licenses principles credit activities and legal consequences.

Also illustrates the hieratical relationships between classes including subclasses that represents specific compliance scenarios and situations. Most of the subclasses are defined through logical restrictions logical rules that bound to them within a specific restricted data set in the ontology and are automatically inferred by the reasoning engine,

By analyzing the ontology graph it's become easier to understand that how the ontology demonstrates and captures the structure of legal framework and how different legal conditions different legal entities interact to determine the compliance outcomes. The visualization therefore prefers a conceptual overview of the ontology design and sports and enhance the explanation of the reasoning processes described earlier in the previous chapters.

By analyzing the ontology graph, it becomes easier to understand how the ontology captures the structure of the legal framework and how different legal conditions interact to determine compliance outcomes.

The visualization therefore provides a conceptual overview of the ontology design and supports the explanation of the reasoning process described in previous chapters.

## 6.5: yEd Diagram

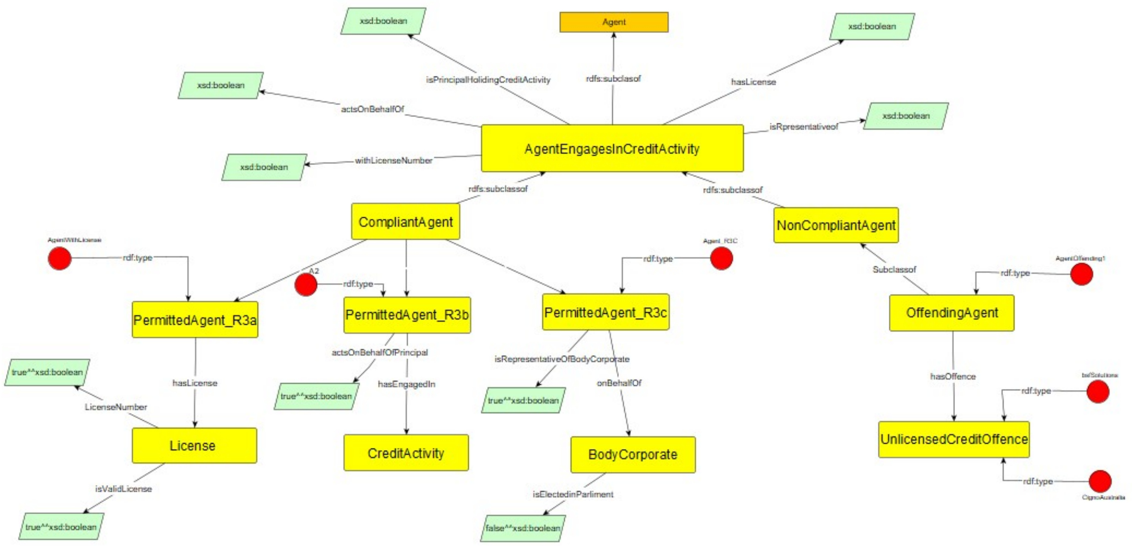


*Figure 7: yEd Diagram showing Visual Representation of the Ontology*

We already discussed earlier about the ontology but above is a pictorial glimpse of our diagram and how it looks like in yEd graph Editor, also I will update up the graph editor file on GitHub so that it can easily be accessible and reproduceable for fixing the issues if occurred.

The Yellow Nodes represent Classes and Subclasses within the ontology, and the main class is AgentEngagingCreditActivity which is subclass of the Agent. The Class further divided into two main categories CompliantAgent, and NonCompliantAgent. These classes satisfies the conditions that the Agent satisfies the legal requirements or not.

The Green Nodes represents the data properties and constraints such as Boolean values, which are used to define the logical conditions in the ontology.

Moreover, the connecting arrows represents the object properties such as hasLicense and actsonBehalfOf etc. which define the relationships between the classes.

Finally, the red nodes represent the individuals' instances in the ontology. These are the specific and generic examples of the agents or entities used or analyzed to test the model and verify they infer and classified correctly by the developed reasoning system.

## 6.6: Chapter Summary

In this chapter presented above we discussed about the visualization of ontology developed in this thesis using the yEd graph editor. The main purpose or the main idea behind this concept or behind visualization of the ontology was to provide to demonstrate a very clear conceptual representation of the ontology structure and to illustrate how the legal concept how the legal entities and how the because friends that are required from the national Consumer Credit Protection Act 2009 or Organized within the ontology model.

The chapter also explained the using of the purpose of using of conceptual diagram to improve the understanding for an even he's not a programmer or developer to better understand the intelligence design. Sometimes we to describe something from concept is difficult to deliver your point or it is sometimes quite challenging for the developers or this the one who is going to explain to describe how it's going how the workflow is working Here comes the solution The YED diagram and that that helps looking for a user to better understand the workflow and the internal relationships between the entities and how the work is done or how the data is retrieving or connecting with each other inside the ontology.

Overall, the visualization supports the ontology design presented in previous chapters and provides an intuitive representation of the knowledge model used for legal compliance checking

# Chapter 7: Ontology Documentation Using Widoco:

Moving forward in this chapter in addition to designing and implementing the ontology There is a need of proper documentation for the ontology and this is very crucial and important part of ontology development. Documentation helps other researchers, Developers and other users understand the struct the concepts and the relationships are defined within the ontology. Without clear documentation it can be quite difficult or it can be quite challenging for others to interprets the ontology or reuse it in different systems that's why here comes the point we use a tool called Widoco for our documentation of Ontology.

And this thesis is the ontology developed for the Representing the legal rules and entities defined in section 29 of NCCP Act 2009 AIR was documented using the Widoco tool. The Widoco is a widely used tool that automatically generates human readable documentation from an ontology file written in OWL. The generated documentation typically includes and contains information about the data of the ontology and the metadata classes properties relationships and visual representation of the ontology structure.

The use of Widoco Allows us give us the permissions and to be presented in a structured standardized and formalized documentation format. This documentation provides a detailed description of the ontology components and allow users the user and developers and others to easily explore the ontology structure. It also improves transparency and makes the intelligence easier to reuse or extent in future research.

## 7.1: Ontology Metadata:

Ontology Metadata provides general information the data about data about the ontology itself. This information describes and demonstrates the purpose and the core scope the authorship and the version details of the ontology defined. Metadata plays an important role in ensuring that the ontology is properly defined and structured and formalized and also model in a way that it can be understood by the other users or systems.

In this thesis metadata information was included in the ontology to describe the context and to formalize Dec structure and purpose of the ontology model. The metadata typically includes and contains elements such as the pathology title description of the ontology and the information about the author and virgin details and some reference to the legal sources used for developing the ontology.

Including metadata in the ontology helps ensure that the knowledge model is properly documented and provides and express the crucial and important background information about the ontology. It also it also supports interoperability and allows researchers end to end users to understand the origin and the purpose of the ontology.

## 7.2: Automatic Document Generation

Although in the presence of multiple advantages of the using widoco one of the main advantages is its ability to automatically generate documentation from the ontology file. Once the ontology is defined in Protégé with all the relevant interconnected relationships and exported OWL format. Widoco can process the ontology file and create a structured formulized documentation website.

They generated documentation includes multiple sections and multiple stacks that descry different aspects of the ontology including ontology description and the class hierarchy object properties and data properties and relationships between the different entities The period of this documentation also help us and provide us a navigable interface that allows users or end to end users to easily express and explore the ontology components.

Automatic documentation generation significantly and reduces the effort required to man document complex ontology spectrum It helps us also to ensure the documentation remains well-structured consistent and with the ontology implementation and can be updated automatically if the ontology is modified and if there is any integration will be updated automatically

## 7.3: Class and Property Documentation:

Despite multiple and great features of the Widoco, one of another important features are the detailed description of ontology classes and properties. The documentation as discussed automatically Lists all the and defined in the oncology along with their parallel relationships.

For each class the documentation provides and demonstrate enough information about its position in the class Haryati and the relationships it has with other classes for example for better understanding let's jump to our classes such as the class credit activity pleasance offense and fans and principal are described in the documentation along with their role in the legal framework.

Similarly, the documentation also describes the other relationships and other things like object properties and data properties defined in the ontology. Object Properties represents relationships between entities such as has license ads on behalf of and has engaged in while data properties stores information an additional information about the individuals about the entities such as lessons numbers or validity status etc.

By providing a structured overview of these elements, the Widoco documentation allows users to better understand how the ontology models the legal concepts derived from the legislation.

## 7.4: Explanation of the Ontology Graph

In this section we will discuss the ontology graph that represents the formalized structure relationships among the key and core concepts of the credit Activities Licensig domain under section 29 of the Australian National Consumer Credit Act 2009.

The main classes that are present includes Agent, Principal, License, CreditActivity, Offence, penalty and Defense. Moreover, there are also some subclasses that contains and express the different legal statuses and roles in licensing compliance.

The ontology graph represents the structured relationships among the key concepts of the Credit Activities Licensing domain under Section 29 of the Australian National Consumer Credit Protection Act 2009.

Object properties and the data properties where We find to represent the relationships between the entities and between the classes that are described in the ontology and the relationship between their attributes. The named individuals were added to describe and illustrate Practical examples of the compliances and offenses. For the better understanding agents that are holding valid Licenses and the other agents that acting on behalf of licensed principal and the agents committing off without a are all represented to test the defined reasoning and compliance logic.

Overall, the ontology graph provides a very simple and an easier visual represent of legal supporting both humans understanding an automated reasoning task. It serves as a connection or as a bridge between the legislative framework and a machine-readable semantic model

Overall, the ontology graph provides a clear, visual representation of legal rules, supporting both human understanding and automated reasoning tasks. It serves as a bridge between the legislative framework and a machine-readable semantic model.

## 7.5: Ontology Documentation Output

The ontology documentation was generated using WIDOCO and how to reproduce this documentation the WIDOCO documentation I will provide the reproducibility steps in our next section. The output provides a user friendly and a very simple HTML presentation of the ontology including the following sections

**Overview**: Summarizes the ontology classes, object properties, and data properties.

- **Class Details**: Lists each class with its hierarchy, equivalent classes, and members (named individuals).
- **Properties**: Provides object and data properties with their domains and ranges.
- **Named Individuals**: Shows all instantiated agents and their property values, illustrating compliance and offence scenarios.

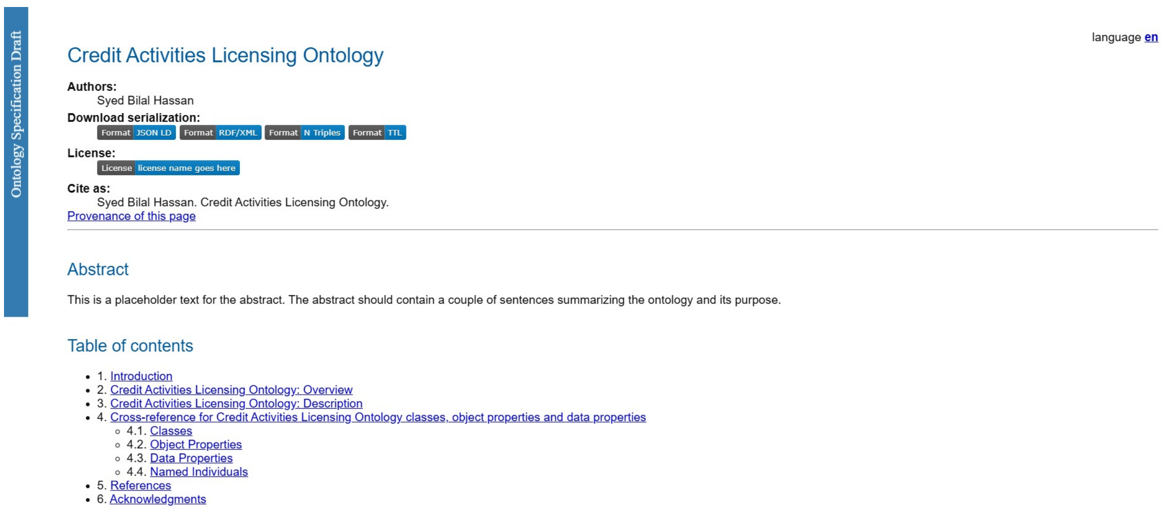


*Figure 8: Widoco Documentation*

Visible screenshot that is attached is showing the Pictorial glimpse of the Widoco Documentation that is generated online. This documentation enables both the human readers and automatic tools to navigate and understand the ontology. It is hosted and available online in the form of fileindex.Html file providing a Single click entry point to explore and navigate the full structure relationships and examples.

By combining the ontology graph and the documentation generated the report ensures transparency clarity and reproducibility of the compliance Creditactivity. Researchers and practitioners can use this documentation to inspect the ontology understand its structure and apply it to automated reasoning tasks.

# 7.6 How to reproduce the Widoco Documentation?

To reproduce document that is donated by Widoco Either not a very difficult task for a non-programmer or practitioner or a researcher you just need to install the widoco and put the widoco jar file Together. It is must keep in mind that both of these files must be in the same folder then you have to run the following command

```
java -jar widoco.jar -ontFile clean.owl -outFolder ./doc -uniteSections
```

and you will find a folder that is created automatically generated in the same location where your ontology is present.

To create this and to run this you must have install and everything is up to date then you open the CMD and go to the folder where your ontology is present and run this command automatically creates another folder in the same location where the ontology is present and then you can navigate to that folder and you will find and file that is named called index.Html and when you open this file you can easily access the document that is created by this automatic tool and you can easily navigate and you can find all the classes and other descriptions and labels over that document.

## 7.7: Chapter Summary

This chapter presented the documentation of the Ontology developed in this thesis using Widoco tool. As we discussed earlier the proper documentation is an important and crucial step in ontology development because it allows other users the end-to-end users and other rese to understand and enhance and formalize the structure and the purpose of the ontology and the components of the ontology effectively and easily.

Initially in this chapter above this chapter first describes the role of ontology metadata and how it provides crucial and required important information about the ontology such as its purposes authorship and Scope. It then Explained automatically this tool generates structured and modeled documentation from The OWL ontology File created in Protégé

Moving forward the chapter also discussed how they generated documentation describes the ontology classes the structure The object proper and the data properties that are crucial and that repress the legal constraints and the legal concepts along with the entities defined in the NCCP Act. The visual graphs generated by this tool also help illustrate the relationship between the ontology elements clearly.

Overall, the documentation using widoco powerful exact and structured representation of the ontology and improves its usability transparency and potential for reuse in future for other semantic web applications by the end-to-end users.

# Chapter 8: Compliance Checking Implementation

This chapter presents the implementation of the companies checking mechanism based on the ontology that will formalized developed in our thesis. The main or primary objective of the system is to demonstrate or you can say is to illustrate how the legal rules and the legal restrictions constraints that defined in the Section 29 of the National Consumer Credit Protection Act 2009 can be represented in a that is quite beneficial in a way add to the machine-readable format and evaluate it automatically using semantic web technologies.

And the compliance checking process relies on several technologies that mostly worked together to sport and demonstrate automatic reasoning and evaluation of legal conditions. The ontology developed in Protege is stored in a semantic database where from that database the semantic database it can be queried and it can be analyzed using a semantic languages and visioning Indians. This approach and this approach in the system allows the system to determine or distinguish whether a particular agent whether a specific and visual or an organization satisfies the legal requirements defined in the legislation.

The implement described in this chapter is focusing on mainly three main components period from these three components one of the first one is the ontology is stored and managed in Graph DB triple store which allows the data to be organized are to be settled in the form of RDF triples. There is another term called Sparql queries that are also used to retrieve and analyze the information stored in the knowledge database set. Finally the OWL reasoning mechanisms I replied in a way so to infer new knowledge and automatically classify individuals and according to the legal rules and legal entities with along the constraints represented in the Ontology.

Together these components foundation of compliance checking system developed in this research

## 8.1: Graph DB triple Store:

In order to perform compliance checking the ontology developed in Protege must be stored in Asman Dictator repository that sports RDF data and semantic queries. For this purpose, ontology was Imported GRAPH DB Which is a widely and most triple store that is designed and modeled for storing and managing RDF based knowledge graphs.

Moving forward or triple store organizes and manages the data in this form of subject predicate and object which represent relationships between the entities in the knowledge base datasets. This structure and this model alliance well with the ontology wear classes individuals and proper are expressed using semantic relationships aging RDF-based knowledge graphs.

Once the ontology was exported from Protégé in OWL format It was then uploaded into Graph Devi as a repository or imported to the Graph DB repository. The repository stores all the required ontology classes properties and individuals as RDF triples. This allows the system to structure and to store structured knowledge and to retrieve or infer relevant information through semantic queries.

The use of graph DB provides multiple advantages especially for the compliance checking systems. It allows efficient storage and retrieval of semantic data it also very advantageous because it sports Sparql curies and integrates reasoning mechanisms that can infer new knowledge from the ontology rules. This makes it possible to evaluate or possible to analyze whether a particular specific agent satisfies the legal requirement defined in the legislation.

In this thesis, the Graph DB repository acts as the central knowledge base that stores the ontology and supports the compliance checking process.

## 8.2: SPARQL QUERIES FOR COMPLIANCE CHECKING

Once the ontology is installed in the triple stored as we mentioned earlier the Sparql queries that are quite effective and beneficial and can be used to retrieve analyze the data contained in the knowledge base. Sparql is a standard query language for RDF data and is widely used in semantic Web Applications.

In the context of this thesis Sparql queries almost used to evaluate and to determine and distinguish whether specific agents satisfy the conditions required for legal compliance. The queries examine relationships between the individuals such as within an adult holds a valid license or whether the agent has engaged in regulated credit activity in the financial market.

For understanding in a better way, let me tell you with an example that if a Sparql query can retrieve agents who are connected to a valid and through the has lessons property Similarly queries can identify and distinguish the individuals or the agents who have engaged in the credit activities without possessing the required authorization defined in the NCCP Act. These queries allow the system to analyze the knowledge as stored in the ontology identify possible compliance or violation scenarios.

Sparql queries also with the above-mentioned benefits is also can be used to retrieve information about offenses penalties and legal defenses associated with specific agents. By combining different kore query conditions, it becomes quite possible to evaluate or illustrate complex compliance situations based on the legal rules and the rest represented in the ontology. The use of therefore a flexible and powerful mechanism for interacting with the semantic knowledge based on performing automated compliance check.

We will add a small example here this shows that how the it retrieves All Agents that are compliant in our ontology.

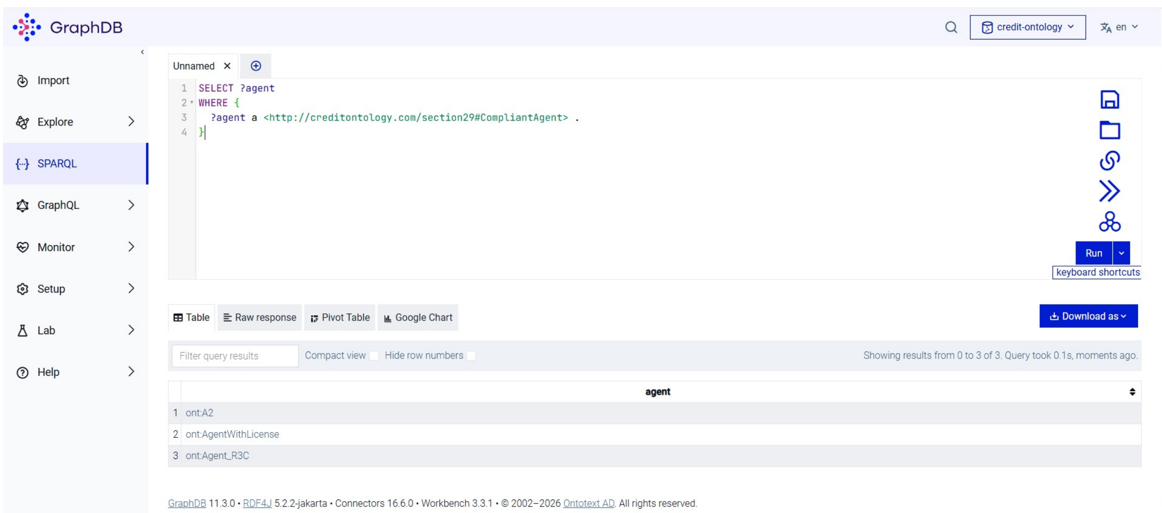


*Figure 9: Pictorial Glimpse of Graph DB and SPARQL query*

We used a simple example in the above screen short that shows that how it works during the Graph DB and how it retrieves information with help of sparql queries, following are the other useful queries that we used in our reasoning system.

I add the basic simple sparql queries in a separate notepad file and uploaded to GitHub and also put all the commands and required sparql queries that are inferring properly so that if someone want to run these queries, he can just install graph DB and import the ontology and can practice these results. The link to GitHub repository is attached below it also contains all the relative material that is required for our thesis like the Ontology file and yEd Diagram and every other thing

https://github.com/SyedBilal61/legal-ontology-credit-licensing

## 8.3: Reasoning With OWL:

In addition to querying the data that is stored in the triple store, there is a reliability on which the compliance checking system will be realized that is O double UL reasoning that is to infer new knowledge from the ontology structure. Reasoning allows the system and make it unable to automatically classify and visuals and the entity is based on the logical restrictions and constraints defined in the ontology.

The ontology that is developed in this thesis also includes several logical conditions that represent legal rules and sections from the Section 29 of the NCCP Act 2009. Most of the circumstances are defined by these rules under which an individual or an agent may legally engage or involve in a credit activity or credit activities be considered in violation of the legislation.

Using OWL reasoning engines such as Hermit and Pellet the system can I or determine these logical restrictions and have the ability to automatically determine the classification of individuals. For example, if an agent is connected to a valid license and it engages or involved in regulated credit activity within the legal framework the reasoning engine may classify as compliant.

On the other hand, if we move forward and check if an agent engages in credit activities without holding a valid license and without satisfying any defense conditions so the reasoning engine or this system may retrieve this or infer that the agent violates the legal requirements defined in the ontology and may lead to the offense and have some penalty.

This is an also fact that the reasoning process allows the system to drive conclusions and automatically without requiring manual evaluation of each scenario. This is the point that shows that the capability demonstrates and illustrates how semantic technologies can support automated legal compliance checking. N

Overall, the combination of ontology modeling, triple store storage, SPARQL querying, and OWL reasoning forms the foundation of the compliance checking implementation presented in this thesis

## 8.4: Chapter Summary

In this chapter we studied and described the implementation of the compliance checking mechanism that is developed in this thesis. The ontology created in Protégé and was stored in the graph DB triple store along the information the knowledge to be organized and structured as RDF triples and managed within a semantic database.

As Parker queries were used to retrieve and analyze the information and the knowledge that is stored in the ontology, this enable this system to examine or express and give the ability to determine whether urgent satisfies the legal requirements defined in the section 29 of the NCCP Act 2009 idiot in addition To WL resin was applied to infer new knowledge and automatically classify the agents and individuals or organization based on the logical rules represented or assigned in the ontology.

The combination of Graph DB, SPARQL queries, and OWL reasoning demonstrates how semantic web technologies can support automated legal compliance checking and provide a structured framework for evaluating legal rules in a machine-readable form

# Chapter 9: User Interface Implementation

This chapter plays a crucial and important role in our umm because this chapter represents the implementation of the user interface developed for the compliance checking system. The user interface is another representation scenario for the representing the data but it is also very appealing and quite helpful for the end-to-end users who are not developers or non-programmers to understand the ontology to understand the workflow how it works. And while the ontology and reasoning mechanism and scenarios provide and determines the underlying legal logic A user interface at that time is must require to allow users to interact with the system in a simple and accessible way.

This interface was implemented as a lightweight or web application using standard web technologies including the HTML CSS and JavaScript. The main purpose and the main objective of the interface is to allow users to enter relevant information about a credit activity scenario and receive an immediate indication or result of whether the situation complies or allies with the legal requirements defined in the section 29 of the national consumer Credit Protection Act 2009.

This system also follows a human centered approach in which legal questions are translated into simple input fields and guided questions. The information over the input data that is entered by the end-to-end user is processed by JavaScript logic that evaluates the compliance conditions and displays the results directly on the interface.

Following sections describes the main components of the user interface including the interface design the HTML for used for user input the JavaScript logic to evaluate compliance conditions and the presentation of the results and the visual ontology diagram created using yEd.

The link to the user interface that we created in our reported can be accessible on the GitHub via the following link.

https://github.com/SyedBilal61/Credit-ontology-UI

and I also created the GitHub webpage which also directly show the required page and you can practice and check the compliance results directly on this useful link which is mentioned below

https://syedbilal61.github.io/Credit-ontology-UI/index.html so, someone can directly access to that above link with import and cloning the whole file and can easily access webpage and that we access and check the compliance results. Follow is the home page of our UI and we can click on the navigation bar and can click one the button and check our compliance results also we can check the compliance result by putting the data in the input fields. Pictorial glimpse of our UI is as follows.


*Figure 10. UI Interface of Our Website*

## 9.1: Interface Design

The user interface was designed with the objectives and purposes of simplicity clarity and ease of use. Since the system intended or designed to demonstrate or illustrate legal compliance checking it is important on the other hand and that also the users without technical knowledge can inter with the interface and understand the results.

The interface consists of multiple web pages that are connected through a navigation bar. The navigation bar provides access to the main components of the application including the compliance checker the detailed compliance form that contains the input and an additional age and form describe being the project and its contributors.

The main page introduces the project and provides an overview of the complying checking case study related to the section 29 of the National Consumer Credit Protection at 2009. At the interface also includes and have some responsive layer that adapts and can be used on different screen sizes ensuring accessibility across different devices and different operating systems.

And styling and layout were implemented using the CSS. The design uses a clean layout with the structured containers responsive form elements and visual indicators that help user interpret compliance results. The main aim of the UI or using the CSS is just to visualize our case study in the form of and also and also do very less logical reasoning on a website because it will take us to out of our goal of thesis. Color coding is used to highlight the outcome of the compliance check, such as green for compliant situations and red for non-compliant situations.

This design approach ensures that the system remains easy to navigate while clearly presenting the compliance checking functionality.

## 9.2: HTML Form for User Input

The main interaction between the user and the complies checking system the UI interface The main interaction occurs through HTML forms. These forms collect information that corresponds to the legal conditions defined in the ontology and legislation

The compliance form or the input form includes several input fields where the user or individual who is interacting with this user interface and details about a potential credit activity scenario. Examples of information collected through the form include the type of credit activity in the financial scenario and the identity of the acting entity and the lessons information and the relationship between the agent and a particular principal entity.

There are some also additional fields that allow users to specify whether there were fees were charged or whether the agent is acting on the behalf of some principal or on the behalf of some director or entity and where there is legal exemption may apply to that agent. These inputs correspond to the legal conditions and logical reasoning described in the section 29 of the NCCP Act.

In addition to the detailed form the system is capable of also providing a simplified question-based interface with some drop down or with some Boolean extractions yes or no. This interface represents the legal conditions as a series of yes or no questions such as whether the user engages in credit activity or whether the user has a valid lesson or rather, he is exempted from a principle or whether he has some sort of lessons on which he is acting upon. This approach allows users to quickly evaluate a scenario and examine a condition without entering extensive data.

By translating legal conditions into structured form inputs, the system makes it possible for users to interact with the compliance checking mechanism in an intuitive and accessible manner.

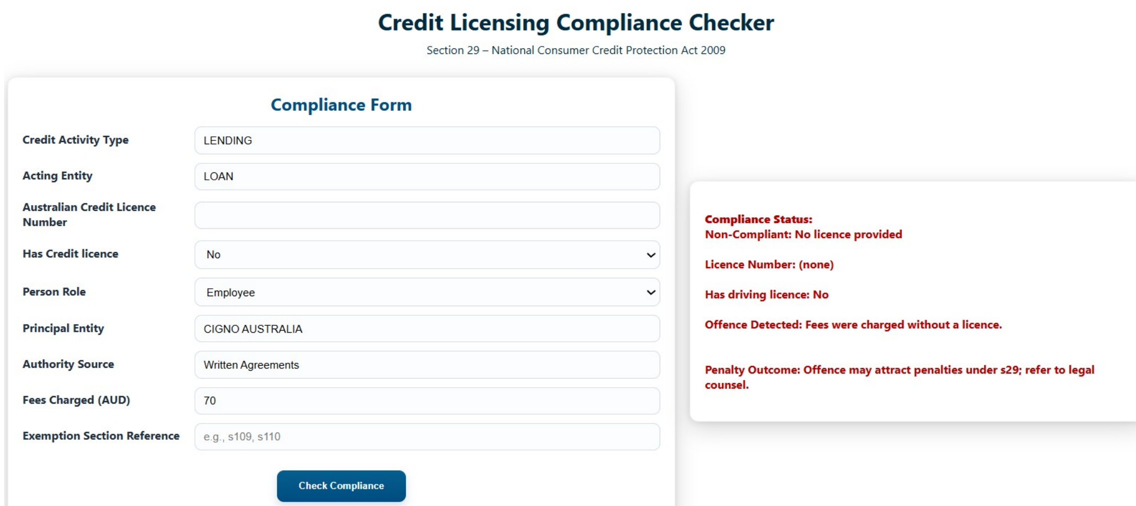


*Figure: 11. Pictorial glimpse of the UI compliance checking input form*

## 9.3: JavaScript Logic for query Execution

The compliance checking logic of the interface is implemented in using JavaScript. One the user submits the form or click submit the button the system corrects the input values or the data that we provide it and evaluates them using conditional rules that represents the logical use of this system or the legal logic of the system.

The JavaScript function responsible for evaluating or for determining compliance reads the values and entered by the user and checks whether specified legal logical condition or met and satisfied for example the logic verifies whether the user corrected activity and whether the user holds a valid Australian Credit license.

Moreover, if the user holds a valid license and possessing the valid make the system identifies this scenario as compliant agent. If the user or the entity does not hold a lesson The system checks whether the activity is performed on behalf of principal ENT and if performed whether that principle holds a valid license or it will check another condition that it qualifies for a legal exemption. N

The logic also valuates whether some charges or fees have been charged without a license or which may indicate a potential offense under the legislation. Based on these current circumstances and conditions the system determines whether the scenario is compliant or non-compliant or whether a legal defense may apply.

Although the reasoning process is simplified in the interface, it reflects the legal structure represented in the ontology and demonstrates how legal rules can be translated into executable logic for automated evaluation.

## 9.4 Display of Compliance Results

Moving forward after evaluating the input conditions the system displays the compliance results directly on the user interface. These result section of the interface dynamically up to represent the outcome of the compliance evaluation.

The system uses visual indicators AH to help users and quickly understand and read the results. If the scenario Satisfies or understand the legal requirements the interface displays a compliant status message using green visual indicators. And moreover, if the scenario violates the legal conditions The interface highlights the results using red indicators and displays a message explaining the potential legal issue.

In situations where a legal defense or our friends may the system may displace a separate notification indicating that the users or the entity may be acting under the authority of a licensed or exempt principal. This result is visually Distinguish or determine using a different color indicator to differentiate it from standard compliant or non-compliant outcome strategies. The results section may also provide additional contextual information in the form of text such as the of a lessons license Possible offenses or potential penalties related to that scenario. These results are dynamically displayed end quickly retrieve or readable without the user to reload the page.

This approach ensures that the compliance feedback is presented clearly and immediately after the user submits the information.

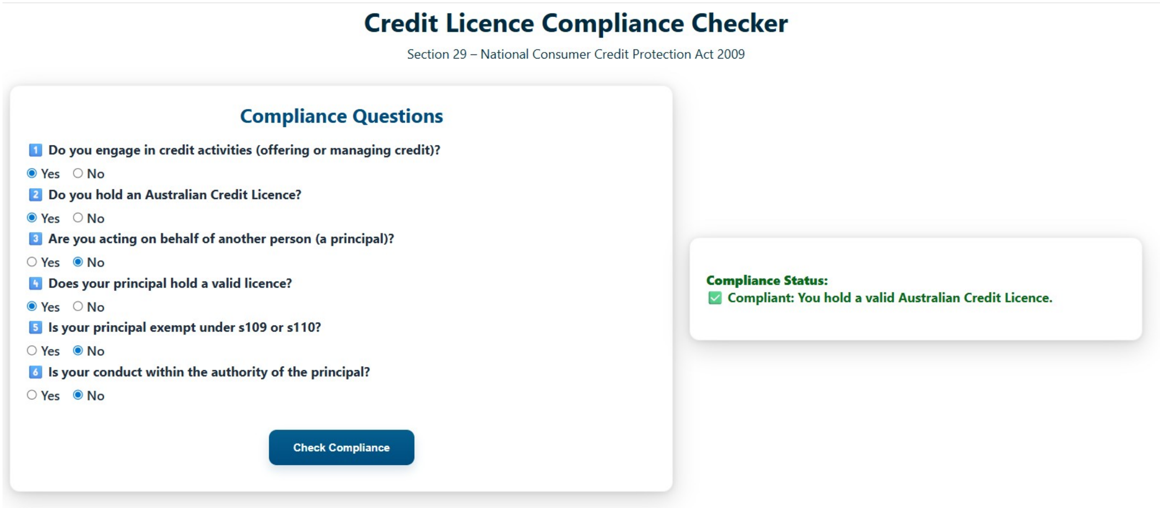


*Figure 12 .UI Compliance checking*

## 9.6: Why a User Interface is Used in the Presence of an Ontology?

Here comes a very quick and very interesting question that why a user interface is used in the presence of an ontology so although the ontology provides the former representation of the legal knowledge and the legal rules and sports automatic reasoning, it is not directly accessible to the most of the users especially those who are nontechnical or non-programmers. Ontology tools such as Protege or triple stores like Graph are designed for technical and require knowledge of semantic web technologies old structures old WL structures add Sparql queries. For non-technical users interacting directly with ontology would be complex and impractical that's why the user interface is used in the presence of an ontology.

As discussed about that reason a web-based user interface was developed to act as an interactional layer between the user and the compliance Based Ontology jacking system.

The interface helps the users and allows these users even if they are non-technical add to input relevant information about the credit activity scenario through simple forms and guided questions. This information is then processed and retrieved by the system which evaluates the condition defined in the ontology and determines the compliance outcome.

The user interface therefore Improves accessibility and with better usability by translating the complex legal and semantic roles into a simple human readable infraction format. It enables users to perform the basic compliance checks if you don't require or without understanding the dense knowledge of ontology or the modeling or semantic query languages.

In this way and in this method the interface acts as a bridge between the underlying ontology-based reasoning system and the end users would need to evaluate legal compliance.

## 9.7 Chapter Summary:

By summarizing this chapter as we discussed earlier it represents of the implementation of the user interface Advert for the compliance checking system, we used multiple technologies stack for the over this implementation for example HTML CSS and JavaScript to provide a simple and interactive and quick accessible way for users to evaluate activity scenarios and Logical Conditions under section 29 of National Consumer Credit Protection Act 2009.

Initially the chapter describes the overall interface that is designed and the navigation structure of the navigation bar of the application. It then and demonstrates how the HTML forms were used to collect the data from the end-to-end users Related to credit activity is the lesson stressing and principal agent relationships their license number or some other specific scenarios as described in the NCCP Act. JavaScript logic then was used to process the input values and our weights and then determines the complies conditions and then based on these legal rules represented in the legal system give us the results in the form of the text.

The results of the compliance checking are quickly accessible by clicking the submit button and without refreshing the web page and they are displayed dynamically within the interface allow users to quickly understand whether a scenario is compliant whether a condition or agenda is non incompliant or whether a legal defense may apply on a specific condition. In addition, a Visual Ontology diagram created using yet was also included illustrate the structure of the legal ontology developed in the research.

Overall, the user interface provides an accessible layer that connects users with the underlying compliance checking system and demonstrates how semantic technologies and legal rules can be integrated into a practical web-based application.

# Chapter 10: Case Study - Section 29 Compliance Checking

Moving forward now we will discuss the case study that is also one of the major requirements of our thesis the case study on the Section 29 compliance checking of the National Consumer Credit Protection Act. In our first that 29 of national Protection Act and we developed the ontology and it was and giving the required results properly. Now we want to discuss and we want to check with the Hill time example and I include the case study of the Australian Securities and Investment Commission against 2 credit activities.

This represents a real-world case study or an example that is used to validate or determine the compliance checking system developed in this thesis in the form of ontology and in the form of the user I interface. The case study is based on or is focused on enforcement action that was taken by the Australian Securities and Investments Commission ASIC against most of the companies and most of the individuals or organizations that involved in unlawful credit activities in the financial market.

Furthermore, the main purpose of this chapter is to demonstrate as to illustrate that how the ontology and the implemented system can be used or can be evaluated to represent a real legal scenario an automatically evaluate compliance that is also aligned with the Section 29 of the NCCP Act 2009.

The case study is simplified and translated into structured and formalized modeled scenarios that can be processed or evaluated by the ontology and the user interface that we developed in our thesis. Each scenario reflects a different legal situation a different legal criterion including compliant a non-compliant and some of the cases where the legal def may apply or practiced.

## 10.1 Description of the Case Study

The case study is based on a legal action or a legal practice brought by the ASIC against the parties namely Cigno Australia pvt Ltd and BSF Solutions Pty Ltd. They regulated authority the federal court founded that these entities or organizations involved or engaged in multiple credit activities that were without holding an Australian credit license and charged consumers excessive and prohibited fees by violating the laws and the restrictions that are bounded them to practice by the Australian credit license.

The case involved a significant financial impact where short term loans were provided to the consumers or the users and large amount of fees were charged against these loans without proper authorization and without proper documentation that were required to practice. The court also found that the directors and the main stakeholders of the company the main member of the companies were intentionally involved in these unlawful activities.

To check the compliance or check the compliance of our ontology in our thesis this case was directly relevant to Section 20 of the NCCP Act 2009. That law prohibits engaging in credit activities Without a Valid license. Therefore, it provides suitable real-world example to test the ontology best compliance checking system.

For the better understanding and for the purpose of the thesis the case has been simplified into multiple scenarios that represents key legal conditions and the core logical Rules such as licensed activity unlicensed activity and acting on behalf of a principal.

## 10.2 Scenario 1: Licensed Agent Performing Activity:

In this scenario an agent engages in credit activities while holding or possessing an Australian Credit License. According to the legal framework and structure that is defined in the section 29 such activity is permitted as long as the agent is properly licensed.

When this scenario is represented or established in the ontology the agent that is associated with a license that is a valid license and linked to a credit activity. System evaluates and determine these conditions and classifieds and infer that agent is compliant.

Similarly on the other hand when this same scenario is also tested through the user interface that we Developed for our ontology the System identifies that a lesson is present and returns a compliant result in green color in the form of the text without loading the page. This scenario demonstrates the correct and lawful execution of the financial credit activities under the Legal framework and legal structure.

## 10.3 Scenario 2: Unlicensed Agent Performing Activity:

Moving forward in our case There is another scenario that represents a scenario that is similar to the real case in which where an agent engages a credit activity in the financial market without holding a valid license that is authorized to them or give them the permission to do the credit activities. Additionally, the agent fees to consumers or the other persons who are getting loans from companies which is considered a serious violation under the legislation.

In the ontology this scenario or this framework is modeled and formalized by linking the agent to a credit activity without associating a valid license. The absence of a combined with the presence of financial charges leads the reasoning system to classify the agent as a violator as a non-compliant agent.

Moreover, on the other hand again one tested through the user interface the system detects that no lessons are provided and that fees are charged as a result the system identifies and evaluates that this situation occurs under the non-compliant model and indicates that there is the possibility of an offense.

In this scenario reflects the core violation and identified that in the ASIC case and demonstrates how the system can detect unlawful behavior from legal entities.

## 10.4: Scenario 3 An Agent on Acting Behalf of a principal

In this scenario from our case study the agent does not hold a license a valid license directly but involved or performing credit activities on behalf of a principal entity that holds a valid license. According to the legal framework that is described it is clearly stated that this situation may be permitted if the If the acts within the authority of the licensed principal.

In to ontology this relationship is represented using the properties that link the agent to the principal entity and specify or mention the authority under which the agent operates. The reasoning system evaluates and illustrates these relationships and determines and distinguish that whether the conditions for permitted activity are satisfied or not.

When evaluated the user interface also for this section the system check whether the agent is acting on behalf of principal whether the principle is lessons and whether the activity falls within the granted authority. If all conditions are met all these scenarios are satisfied then the reasoning system classified as compliant agent.

This scenario demonstrates how indirect authorization can allow lawful participation in credit activities.

## 10.5: Scenario 4 An Agent with the Legal Defense

This is another that is representing A case where an urgent may not fully satisfy or meet the standard compliance conditions but the agent can rely on a legal defense here it comes very interesting and challenging for example for our better understanding the agent may be acting under the authority of a principal who is exempt under specific legal provisions or some specific legal conditions.

In the ontology such situations are represented and expressed using defense related classes and properties. These elements allow the system to capture exceptions and special legal conditions and legal scenarios and some legal restrictions that may justify the agent actions.

When processed by the reasoning system or evaluated by the reasoning system the presence of a valid defense prevents them from being classified or being inferred in the non-compliant agent scenario. Similarly in the user interface the system identifies data defense applies and provides a corresponding result including that the activity may be legally justified.

This scenario highlights the importance of modeling not only strict rules but also exceptions and defenses within a legal compliance system.

## 10.6: Results of the Complaint Checking system

The results of the compliance tracking system from both the ends the ontology with the help of the reason and from the user interface demonstrates and explains that the ontology and the user interface are able to correctly and efficiently evaluate the different sections and different legal scenarios based on the rules defined in the section 29.

In all tested scenarios system produce efficient results and represented with the expected legal outcomes. License agents were correctly classified as compliant while unlicensed agents engaging in credit activities were identified and distinguished as non-compliant. Cases involving authorized and legal defenses were also handled and practiced appropriately.

The integration of the ontology Disney and the user interface logic allows and gives the opportunity to the system to stimulate real world legal decision making in a simplified and accessible. Although the implementation is designed for explaining expressing and demonstration purposes but it shows the potential of semantic technologies and in supporting automated legal compliance checking

# Chapter 11: Evaluation and Discussion

This chapter evaluates the proposed ontology-based compliant checking system developed for the Section 29 of the Australian National Consumer Credit Protection Act 2009 along with a case study that also compliant on tracks the reasoning system of our ontology perfectly. It also reflects and shows the effectiveness and of the ontology model that is formalized and highlights the advantages of using a semantic approach for legal compliance for a specific legal section or a legal scenario and discuss the limitations and other restrictions encountered during the implement.

## 11.1 Effectiveness of Ontology Model:

The ontology that is model and formalized and developed in this study proved to be effective in representing and expressing legal knowledge that is related to licensing requirements in credit activities that are performed in the financial markets. By formalizing and modeling the legal concepts such as agent license credit activity and defense the ontology enabled structured and machine interpretable representation of legal rules.

The use of OWL based class Restrictions and logical sections which are constraints allowed the system to automatically and quickly and efficiently Classify agents as compliant or non-compliant based on their properties that are defined. This can be easily understandable for example the and the organization or the agent or the holding a license or acting on behalf of a licensed principal work correctly inferred as compliant. Similarly, agents performing credit activities in the financial markets without proper authorization charges extra fees to the consumers were identified as noncompliant.

The reasoning mechanism demonstrated and described consistent and predictable results across all defined scenarios in the case study of the thesis. This indicates and defined the ontology model aligns Well with the logical structure of section 29 and can effectively support compliance checking.

Overall, the ontology successfully captures and identifies the essential legal conditions and translates them into a form that can be processed automatically, reducing ambiguity and improving clarity.

## 11.2 Advantages of the Ontology-Based Legal Compliance

There are a number of advantages of the ontology-based approach but one of the main advantages of the ontology-based approach is the Formalization of Legal Knowledge. Unlike traditional and other text based legal interpret the ontology help provides well formalized and a structured and precise representation of the rules and the legal and Constraints Which are results in reducing the misinterpretation.

Another key benefit and a core advantage of the ontology-based approach is the automation. The integration of OWL reasoning allows the System to Automatically determine comply without requiring and the necessity of manual legal analysis for each of the case. This can significantly and ultimately improve the efficiency and especially in the large-scale regulatory environments.

The ontology-based approach also ensures one of the key factors and this one is the most important factor is called the consistency period since the same logical rules are applied uniformly and the system produces consistent results across different cases avoiding subjective interpretation.

Moreover, moving forward, the ontology there is another main advantage is its **flexibility and reusability.** It can be extended or adapted to include another additional legal provision are adapted to different jurisdictions with minimal. This makes it a scalable and accessible simpler and easier solution for broader legal compliance applications.

Finally, the use of the tools in our technology stack such as Protege Graph DB and SPARQL Demonstrates that ontology-based systems can be practically and easily implemented using existing technologies.

## 11.3 Limitation of the System

Despite in the presence of many advantages but the proposed system I had several limitations that should also be acknowledged and should also be understandable by the users and also by the developers.

The first and the foremost reason behind the limit is that the ontology has simplifies complex legal language into formal logic. While this is necessary for some of the basic functions or for some of the computational processing but it may not capture all nuances, Exceptions or restrictions All logical reasons or interpretations present in the real legal practice.

Moving forward there is another limitation that should also be considered for discussion is that the system is limited to a specific legal provision for instance a section 29. It does not cover the full complexity of credit law or the sections of the legal rules or related regulations. Expanding or express the structure of the ontology to a broader legal framework to a higher would significant and extra additional effort and it is quite challenging and time taking.

And other one of the most significant and concerning limitation is that it the dependence on accurate data. The correctness or the reliability of the compliance results depends entirely and fully on the information provided to the system. If incorrect our incomplete is or given to the ontology the system may produce misleading conclusions or results that will result in the reliability of the system and take time effects credibility of the system.

Moreover, OWL reasoning operates and function under specific and certain logical constraints and logical restrictions and may not fully express or support more advance legal reasoning concepts such as infusibility or such as interpretation based on context.

Finally in a conclusion making in a nutshell the current implementation focusses on a prototype level system. While its demon feasibility further develop would be required deploy it in real world legal regulatory environments.

# Chapter 12: Reproducibility and Availability

In the above chapters we discussed how we can use ontology and what are the benefits and what are the challenges in approaching or accessing these results, what are the limitations etc.

Although we developed the whole ontology from the scratch, them make the yEd diagram for it and run sparql queries and check that all the discussed classes and and all the required classes are inferring properly.

Later on, I also developed the UI so that we can check the inferring results with the help of Boolean conditions and also with the help of input html form , and check that the results are inferring properly. The main goal was is to check the result although it is not required in our ontology but we made it just for the researchers and practitioners and for the non-technical users so that they can understand properly everything from UI interface.

Here I will describe the section so that if someone want to practice and run and check the reasoning system even, he is non-technical guy so it will be easier for him to understand the process.

1. Open protege and create Ontology and make the relationship between the object properties and data properties
2. Add the labels and comments on each section of the class and on each object properties and data properties.
3. After adding the labels and comments you can add the individuals and also connect them with the help of the object properties and data properties to the classes that are described in our ontology.
4. Now you can run the from the menu bar click on reasoner and then click on your required reasoner I used the hermit but there is also pallet reasoner is available
5. If you defined your classes properly their relationship properly than the individuals that are related and that are defined according to the legal logical conditions will infer properly in this section instances on the right side of the ontology if you click on a class for example if you click on a class called Compliant Agent then individual named agent with license will appear properly and infer end this action.
6. Now our ontology is working properly and inferring the required results Now you can save the ontology and in the form of RDF OWL syntax
7. The next step is to create the diagram with the help of YED editor in which you can create the diagram that is showing the relationships

between the classes and object properties and data properties and their interconnected relationships that how they are connected with each other. The Diagram is available on the link that contains the ontology and other things I will also add the link here below

8. https://github.com/SyedBilal61/legal-ontology-credit-licensing
9. There is a tool called Widoco (lode), This tool is used for generating documentation of the ontology, So the researchers' practitioners and the non-technical users can read the documentation and how they are presented and how they are used the logical conditions in the ontology easily.
10. To use this there are multiple ways but there is a condition that the ontology file and the widoco Jar file both should be in the same folder and you can run these commands by the bash through CMD and then going to folder where these files are present.
11. The Commands for the Documentation Creation of Widoco is java -jar widoco.jar -ontFile clean.owl -outFolder./doc -uniteSections you just open the cmd and go the folder where your ontology and jar file are present and then run this command it will create and index.html in a doc folder you can open the doc folder and then open Html file to see the widoco documentation.
12. After this you need to run SPARQL queries These queries present in the GitHub repository and you can run this and check the outcomes.
13. To run these, you should the software called Graph DB Editor and you should create repository in that folder and later after creating the repository you should import the owl that is your original ontology file and then go queries section and run the queries. The example for a simple query is as follows (*Get all agents*

    *SELECT ?agent*

    *WHERE {*

    *?agent a <http://www.example.org/credit-licensing-ontology#Agent> .*

    *}*)

14. From there is a notepad field called queries Text And you can check all the queries that they are properly and give the desired and expected outcomes.
15. Now here comes the one of the most crucial and important part that is the UI interface of our ontology as described earlier although the UI is something else that that over ontology is going to but it is developed for better understanding and for the understanding of researchers and non-technical users.
16. To do this you use GitHub and clone the following repository

https://github.com/SyedBilal61/Credit-ontology-UI

17. After cloning the repository, you can open any of HTML file are I created the GitHub pages directly so you don't need to clone the GitHub and you can check the web page directly on the web page without refreshing the page so there is a folder on the right side of the GitHub a file name called GitHub Pages and you select it and it will open a you UI for our ontology.
18. There are three sections in the navigation bar 1 that checking for the compliance checking the second is the input form in which we can input our data and check the compliance results and Information is about the researcher and the mentor who created these ontologies.
19. Both of the GitHub repositories links are all the following so you may check and check the required results and check that they are infringing properly if you want to understand it better.
    i. https://github.com/SyedBilal61/Credit-ontology-UI
    ii. https://github.com/SyedBilal61/legal-ontology-credit-licensing

# Chapter 13: Conclusion

This chapter concludes the thesis by expressing and reflecting the overall objectives requirements and the Contributions and outcomes of the research. The study aimed to explore and express how the legal knowledge particularly licensing requirements in credit in the financial markets can be transformed into a formal and machine interpretable structure using ontology-based techniques. By focusing on the section 29 NCCP Act 2000 This work demonstrate and explains how semantic technologies can support the automated compliance checking in a clear consistent and scalable manner that is simply easier and accessible.

The conclusions and the results of this research highlight the potential of ontologies not as a knowledge representation tool but also as a practical mechanism or a forum for bridging the gap between the legal text and the computational systems. Through the integration of the modeling and form and reasoning and querying techniques the thesis provides a complete framework for understanding and evaluating legal compliance.

## 13.1 Summary of the Contributions

This this makes several important and crucial contributions to the field of legal informatics and Semantic Web Applications

Initially it represents the formal design and development of a legal ontology that captures and reflects the core elements of licensing requirements in credit law as it is already discussed above by identifying and analyzing and modeling key concepts such as agent license credit activity principle and defense the study translate Complex Legal provisions the difficult logical reasons and sections that are correlated to logical conditions into a structured and logically consistent format. This transformation or this process reduces ambiguity and enables and opens a more precise interpretation of legal rules.

Secondly, the research explains and demonstrates the use of OWL (Web Ontology Language) to explain and to define formal class restrictions and relationships that reflects the conditions specified in the section 29 idiot these logical definitions The specific logical conditions allow the system to automatically infer whether in that is going to involve in a credit activity Is compliant or non-compliant based on the available information. This represents a significant step a major role toward Automating legal reasoning processing that are traditionally performed manually.

Another one of the major and key contribution is the implementation of a compliance checking framework That combines ontology reasoning with SPARQL querying. The understanding and use of a triple store environment Support efficient storage and retrieval of data while enables us to inter with the system and extract meaningful and required compliance results as that are defined and that are required.

The inclusion of a real world inspired case study also and further strengthens. The practical relevance and the requirement of the work. By expressing and applying the ontology to the different scenarios such as license agents and unlicensed AG and agents acting on behalf of a licensed principal and adjourns with legal defenses as described also earlier the demonstrates its ability and cap to handle and practice a variety and plethora of legal situ accurately. The successful inference of results confirms that's the ontology aligns well with the intended legal interpretation.

In addition, the thesis describes an integrates multiple Tools in the technology stack including Protégé for ontology development Z YED for visualization Graph DB for data management and documentation tools for ontology description. This end-to-end implementation shows that ontology-based compliance systems are not only theoretically but also practically achievable Accessible and approachable using widely available technologies.

## 13.2 Future Improvements

Despite the and Promising results Exciting There are also several opportunities for further next enhancement and expansion of this work for the newly related persons.

One of the more important and key directions for future research is the extension or enhancement of the of the ontology to include a broader range of legal provisions and legal sections within credit law or even across different areas and aspects and scenarios of the legislation and other laws. Expanding and expressing the scope would allow the system to address and express complex and different and difficult regulatory requirements and provide a more comprehensive compliance solution.

Another potential improvement and beneficial assessment Lies in enhancing the reasoning capabilities of the system. While OWL Provides a strong foundation for logical inference. It has limitations when dealing more complex and more advanced legal conditions and concepts such as exceptions conflicting rules interconnected expertise and context dependent interpretations. Incorporating approaches Such as rule based or hybrid reasoning frameworks could improve this ability or credibility model or formalize the new real world legal complexity.

The development a more advanced and user-friendly web interface Is also highly recommend. Such an interface could enable legal practitioners, Regulators Our business users or from Organize to end data easily and visualize results understands the outcomes and understand compliance without requiring technical knowledge of the ontologies or query languages.

Furthermore, in addition to this integrating the system with the real data sets from the knowledge base datasets and regulatory databases that are formalized and modeled would significantly enhance its practical applicability. This would allow the ontology to be tested in real compliance monitoring scenario and provide reliable and consistent and accessible easy dynamic results.

Performance and scalability or additional aspects that should be considered, especially if the system is to be deployed in a large scale environ. Optimizing And expecting query perform and organ large volumes of data will be essential for real world adoption.

## 13.3: Final Remarks

In conclusion this thesis demonstrates and express that the ontology-based approach offers a power and promising an exciting solution for modeling legal knowledge and automating compliance checking. By transforming legal text or legal conditions Logical legal rules into formal representations it becomes possible to achieve a higher level of clarity and consistency and efficiency in legal analysis in a very simpler easier and accessible way.

While the complexity of legal reasoning cannot be fully focused, captured by current technologies, The results and the outcomes of this study shows that significant progress can be made and can be achieved through the use of Semantic web tools and structured knowledge models. The ability and the capability to automatically read and infer compliance outcomes highlights the practical value of in both academic and real word contexts.

This work represents a foundational and a fundamental approach toward the development of intel and quick legal that can Help and assist in regulatory compliance, Decision-making, and legal analysis. With further research and development such systems have the ability and potential to analyze and support a wide range of applications that are contributing to the advancement of digital law and legal informatics

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## اظہارِ تشکر:

ابتداء میں یہ صرف ایک خواب تھا جس سے شروعات ہوئی تھی۔ سفر شروع کیا تو ہر قدم کے ساتھ راستہ روشن ہوتا گیا اور آج یہاں ہو کہ میں نے اپنے تعلیمی سفر کی ایک اور منزل طے کر لی۔ میں اپنے **اللہ عزوجل** کے کرم کا بے حد ممنون و شکر گزار ہوں جو لمحہ بہ لمحہ میرا رہبر و ہادی ہے اللہ کی ذات کے بعد میری **ماں** کی دعائیں ساتھ نہ ہوتی تو شاید اس کامیابی کا حصول ممکن نہ تھا یہ کامیابی ان بالوں کے نام جو ہماری فکروں میں سفید ہو گئے اور جھریوں والے ہاتھوں اور کپکپاتے دعا گو ہونٹوں کے کے نام جو ہمارے کامیابی کے لئے ہم تن دعا گو رہے **والد سید اعجاز حسین (مرحوم)** اگر وہ آج زندہ ہوتے یقیناً خوشی اور فخر محسوس کرتے ہر کامیابی خراج مانگتی ہے۔ میرے اس سفر کی قربانی ماں کے حضور سے دوری تھی۔

عمر میں چھوٹے اور رتبے اور فیض میں بڑے، چھوٹے بھائی **سید زکی الحسن** کے نام جو ہمیشہ میرا دستِ بازو بنا اور جس کے شانہ بہ شانہ ساتھ نے مشکل سفر کو سہل بنایا۔ میری زندگی کے روشن ستون، **میرا بھائی اور بہنوں کا شکریہ**، جن کے پیار، تعاون، حوصلہ اور محبت نے ہر مشکل سفر کو آسان بنایا۔ آپ کے ساتھ کی گئی باتیں، ہنسی خوشی کے لمحات، اور ہر قدم پر دیا گیا سہارا میرے لیے بیش قیمت ہے۔ آپ کی محبت اور قربانیوں نے مجھے ہمیشہ آگے بڑھنے کی ترغیب دی اور میری کامیابی کو ممکن بنایا ان کی محبت میرا قیمتی اثاثہ ہے جس کا تشکر کبھی لفظوں میں ادا نہیں کیا جا سکتا۔

میری زندگی کے سب سے انمول اور قیمتی تحفے، میری شریکِ حیات، **سرتاج**، اور میرا بیٹا، **جبران**، جن کی محبت، مسکراہٹ اور موجودگی نے میری دنیا کو روشنی، خوشیوں، محبت اور ایک مقصد سے بھر دیا، اور جن کی وجہ سے ہر لمحہ قیمتی اور ہر کامیابی حقیقی معنوں میں مکمل محسوس ہوتی ہے۔ میری زندگی کا حصہ ہونے اور خوشیاں بکھیرنے پر میں ان کا بھی شکریہ ادا کرتا ہوں مالک ہمارا ساتھ سلامت رکھے (آمین)۔

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**سید بلال حسن**

**ماسٹرز ان کمپیوٹر سائنس**

**یونیورسٹی آف فلورنس، اٹلی**

# LAST BUT NOT THE LEAST

Last but not least, I wanna thank me…

I wanna thank me for believing in me

I wanna thank me for doing all this hard work

I wanna thank me for having no days off

I wanna thank me for never quitting

I wanna thank me for always being a giver and try to give more than I receive

I wanna thank me for trying to do more right than wrong

I wanna thank me for just being me at all times

# Useful Links

## UI Website Interface

https://github.com/SyedBilal61/Credit-ontology-UI

**Ontology and yEd diagram and SPARQL queries**

https://github.com/SyedBilal61/legal-ontology-credit-licensing

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